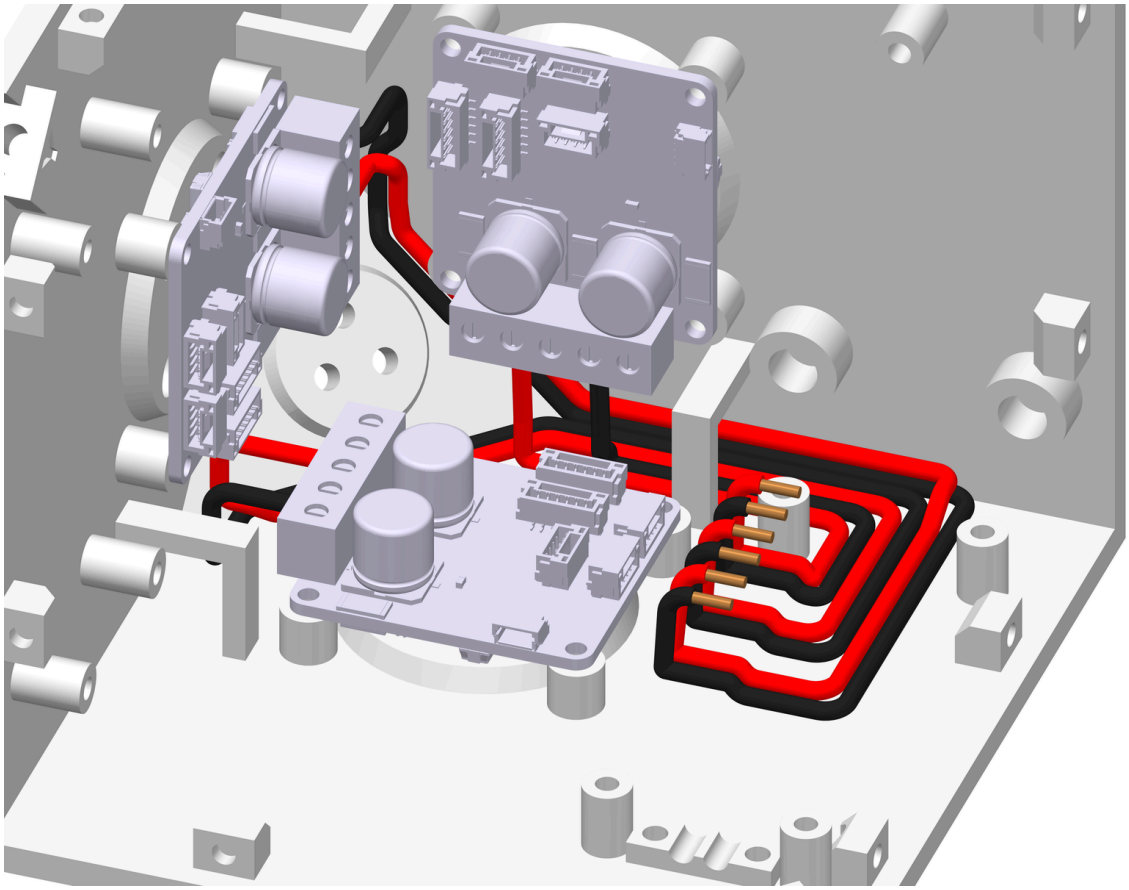
Wiring

Every wire, in order.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only

One thing per page

One action, one big picture. Short sentences, much white space.

Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).

Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).

For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.

Print it right

In color, at 100 % - not 'fit to page'.

Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first

Grown-up nearby

Work with your teacher or mentor in the room.

Battery last

The battery (LiPo 6S, 22.2 V) goes in LAST - only after the checks at the end of this guide. Never let its two leads touch: it burns.

A damaged battery

A battery that is puffy, hot or damaged is not used. Tell your teacher.

Hot!

The soldering iron and the heat gun are hot. Back in the stand every time.

Fresh air

Solder smoke is not good to breathe. Open a window.

Eyes

Glasses on when you cut wire: the ends fly.

Before you start

The motors are in

The 3 motors with their motor drivers are in the box.

The boards are not

They go on at the end, over the wires.

The clamps are printed

clamp_z, clamp_y, clamp_x_1, clamp_x_2, and their 8 spacers.

The order

- 1 Motor power - the 6 thick wires on the floor
- 2 Battery supply - fuse, battery connector
- 3 Board power - the 5 V wires
- 4 Signals - links, CAN, I2C
- 5 Clamps on, boards on, every wire into its terminal, check

Why this order? What lies lowest goes in first. Every wire then lies on the ones under it, never under them.

You need

Wire stripper and side cutter

No ferrules: the terminals have a clamp plate. Stranded ends: strip 7 mm, twist tight - no solder on them.

Crimp tool

For insulated terminals (the blue size, 1.5 - 2.5 mm2) - the fuse lugs.

Soldering iron, solder

And a heat gun with heat-shrink tube.

Small flat screwdriver

For the terminals.

Masking tape and a pen

A name flag on every wire; a permanent marker for the cut mark on every wire end.

Multimeter

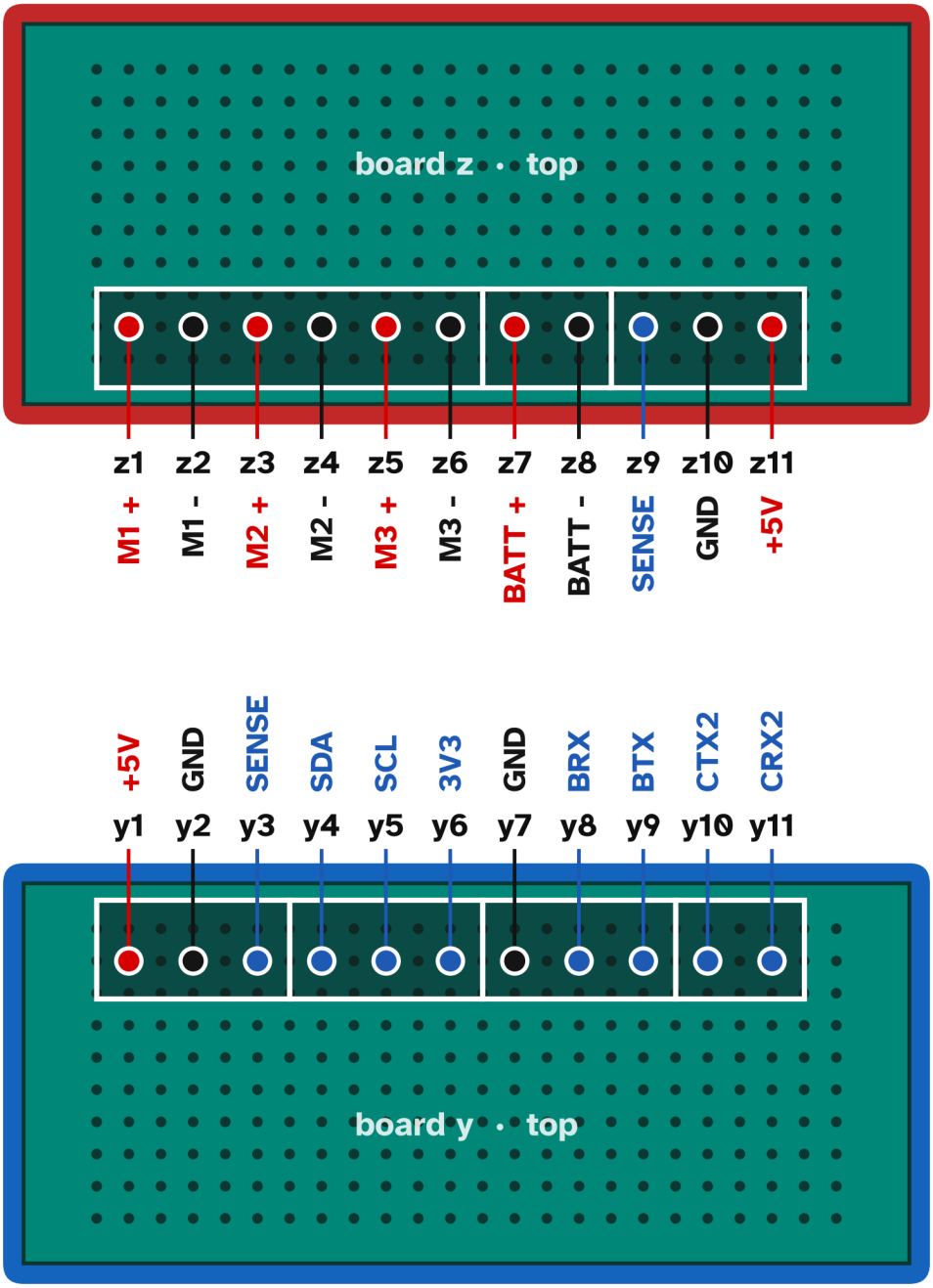
For the checks.

The 4 cable clamps

And their 8 spacers - no screws.

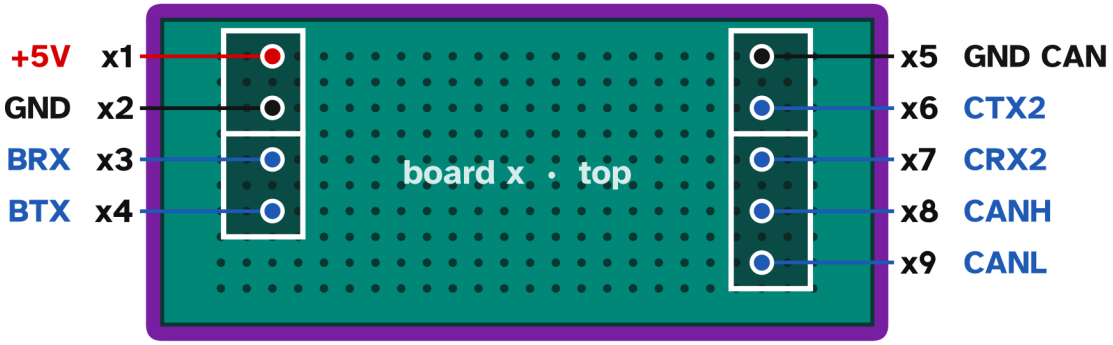
The terminals

Each pole has a name: the board and a number - z7 is pole 7 of board Z. Every step uses these names.



The terminals

Board X: two blocks, one on each side.

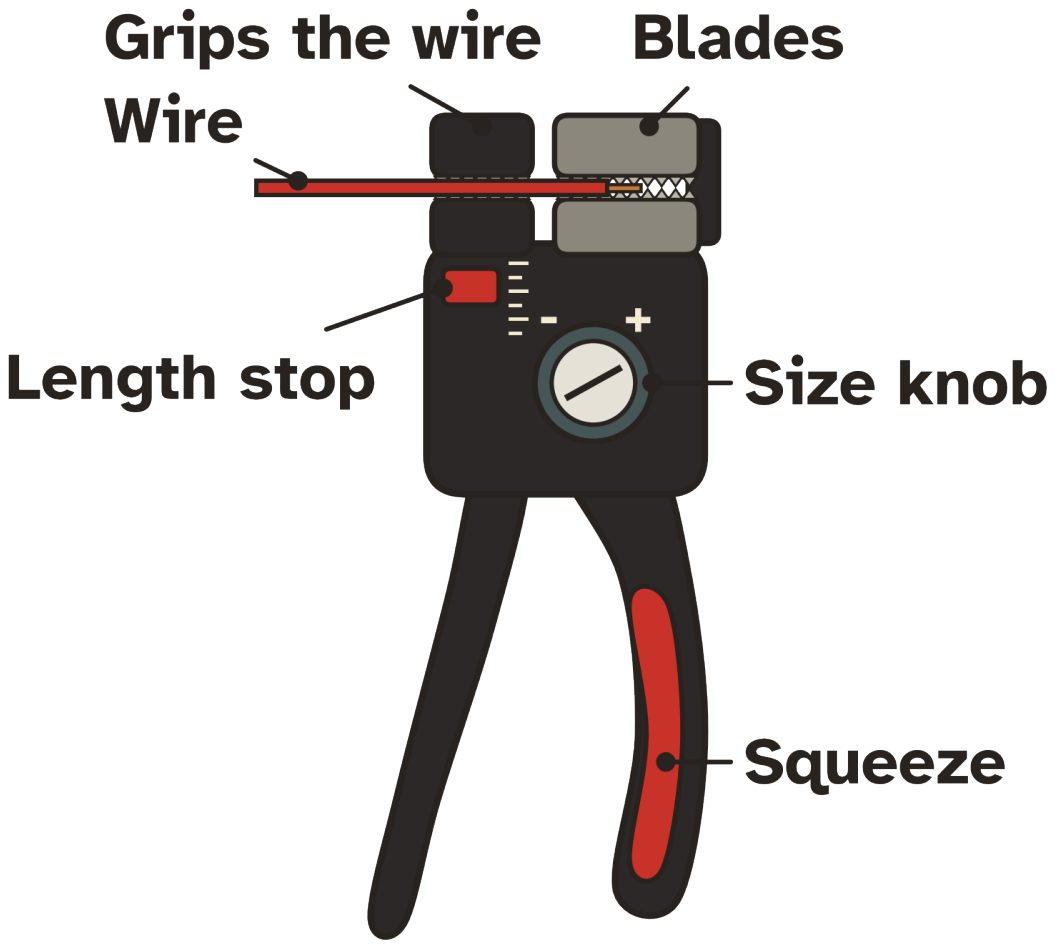


Pole to pole

- **z9** -> y3 SENSE
- **z10** -> y2, x2 GND
- **z11** -> y1, x1 +5V
- **y1** -> z11 +5V
- **y2** -> z10 GND
- **y8** -> x3 BRX
- **y9** -> x4 BTX
- **y10** -> x6 CTX2
- **y11** -> x7 CRX2
- **x1** -> z11 +5V
- **x2** -> z10 GND

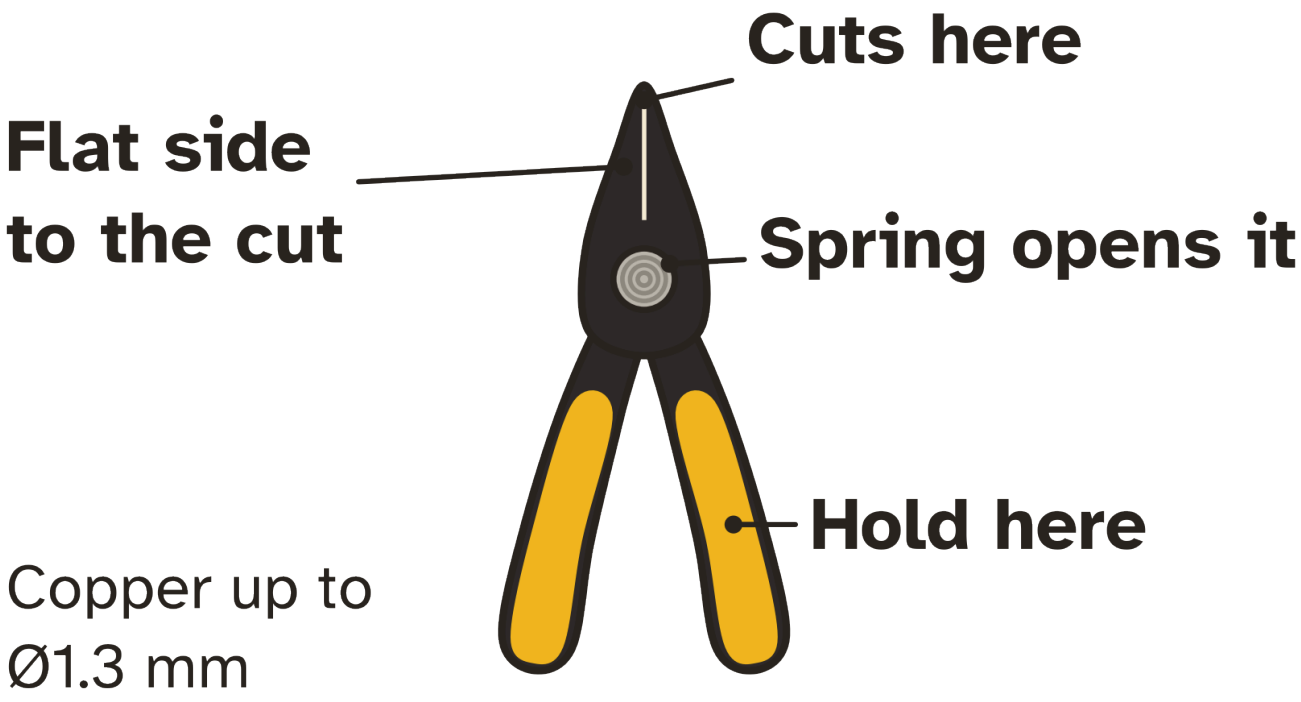
Your tools: the wire stripper

For every wire end.



- Set the knob to the wire size. Put the wire in up to the length stop. Squeeze.

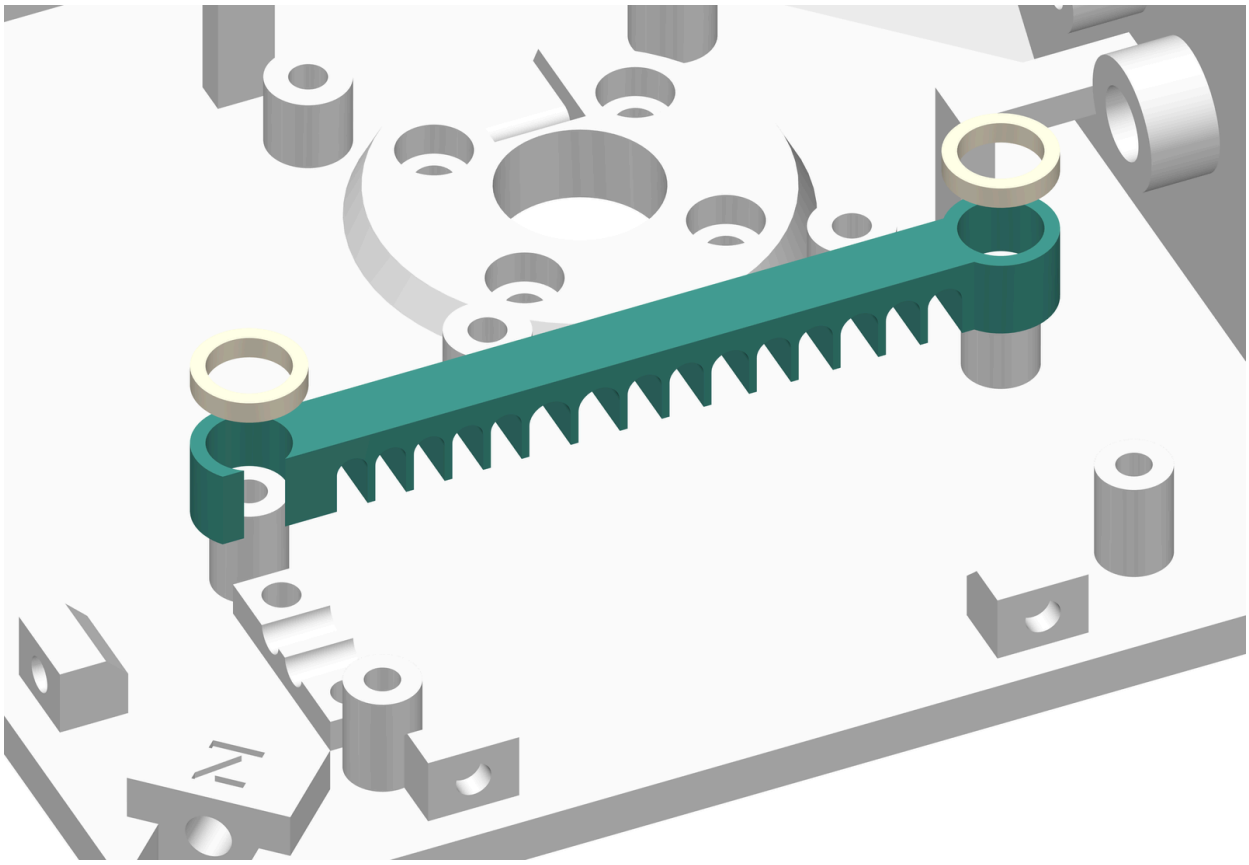
Your tools: the side cutter



- The flat side toward the cut. Only copper up to 1.3 mm.

The cable clamps

A bar under the edge of each board - two at board X. Every wire has its own arch: the wires stay in order and do not move.



- The wires go in first, on the floor. The clamps go on at the end, OVER them: each arch over its wire, the rings down over the board standoffs.
- A ready-made cable (CAN, I2C) goes under one round arch. At clamp_x_1 the wires passing inward share one wide arch.
- A spacer on each ring, then the board. The board presses on the spacers and holds the clamp down.

Cut list · motor power

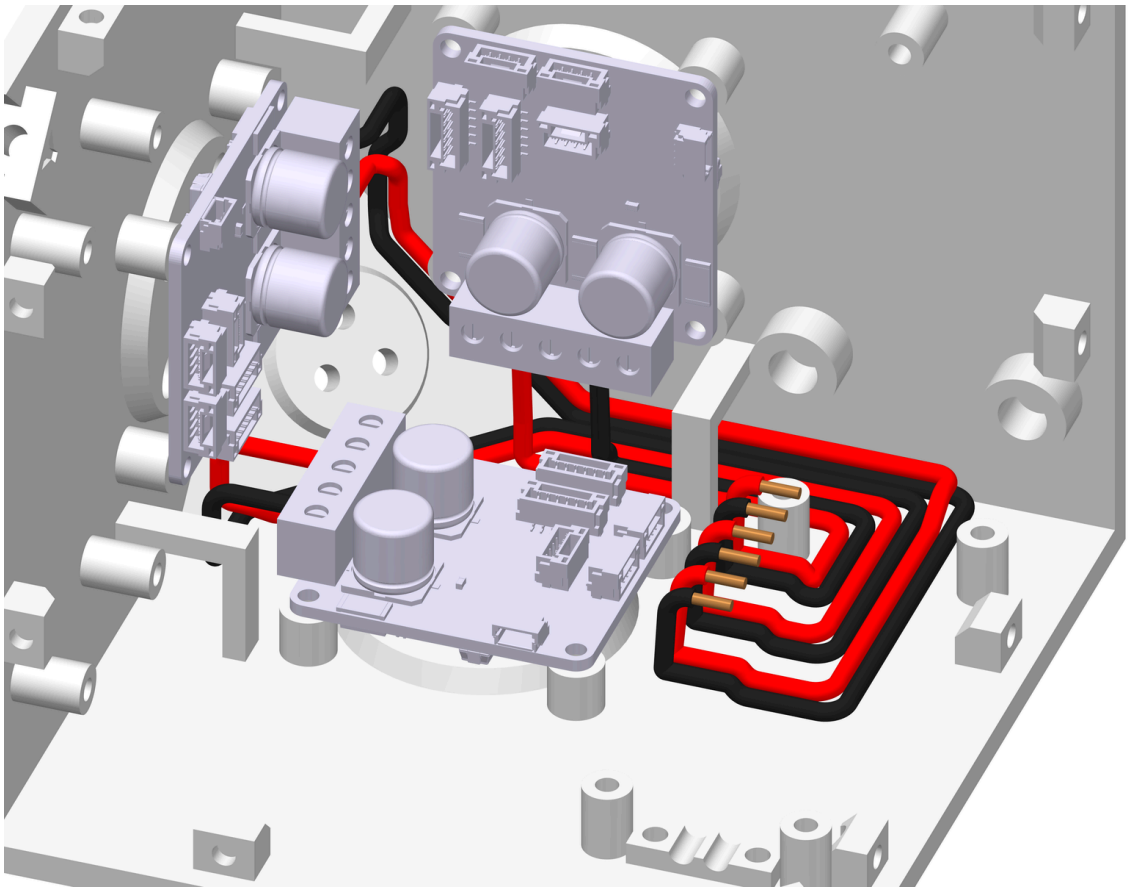
1.5 mm2 silicone wire. Length = route + 10 mm.

wire	color	from	->	to	cut
M3-	black	driver Y	->	board Z, z6	220 mm
M3+	red	driver Y	->	board Z, z5	200 mm
M2-	black	driver Z	->	board Z, z4	200 mm
M2+	red	driver Z	->	board Z, z3	180 mm
M1-	black	driver X	->	board Z, z2	110 mm
M1+	red	driver X	->	board Z, z1	105 mm

- Strip 7 mm at both ends and twist the strands tight. No ferrule, no solder.
- Put a name flag (tape) on each wire: M3- , M3+ ...

1 · Motor power

6 thick wires: from each motor driver (+ and -) to board Z, z1 to z6. They lie on the floor, side by side.

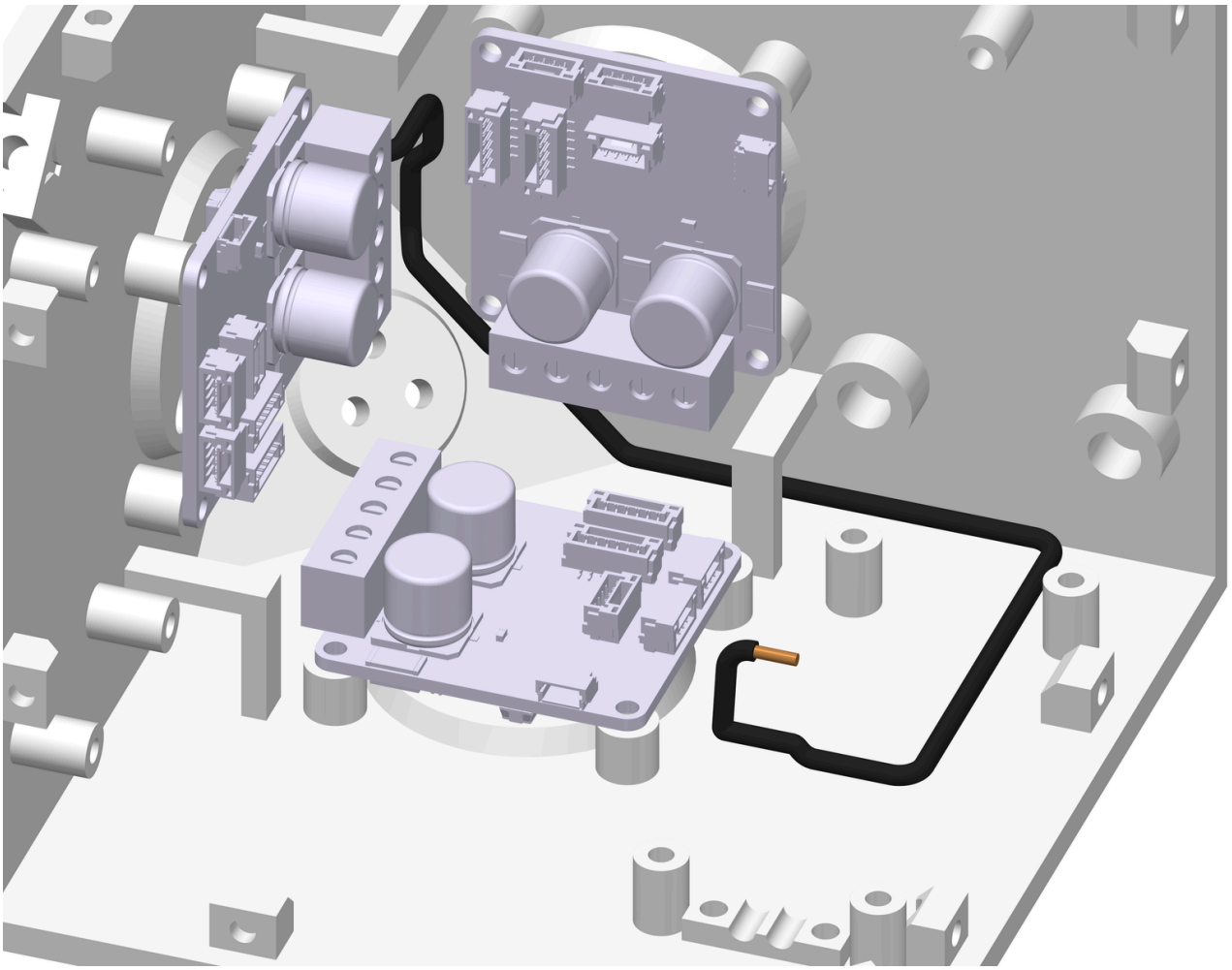


The 4 clamp bases on their standoffs first.

Why?

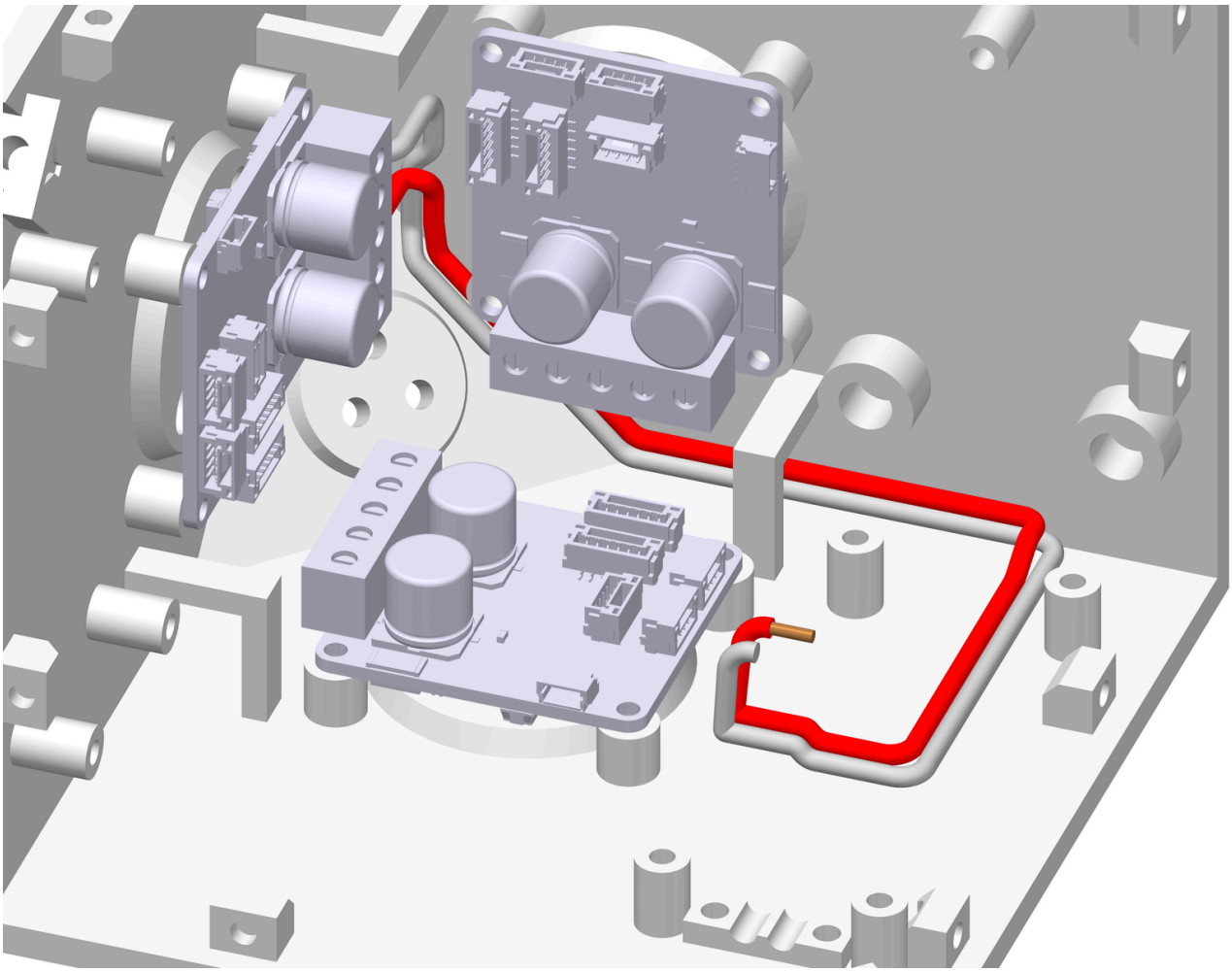
When a motor speeds up its wheel it pulls a lot of current - many amperes. A thin wire would get hot. Board Z takes the battery power and shares it out: each motor driver gets its own pair, + (red) and - (black).

M3- · black



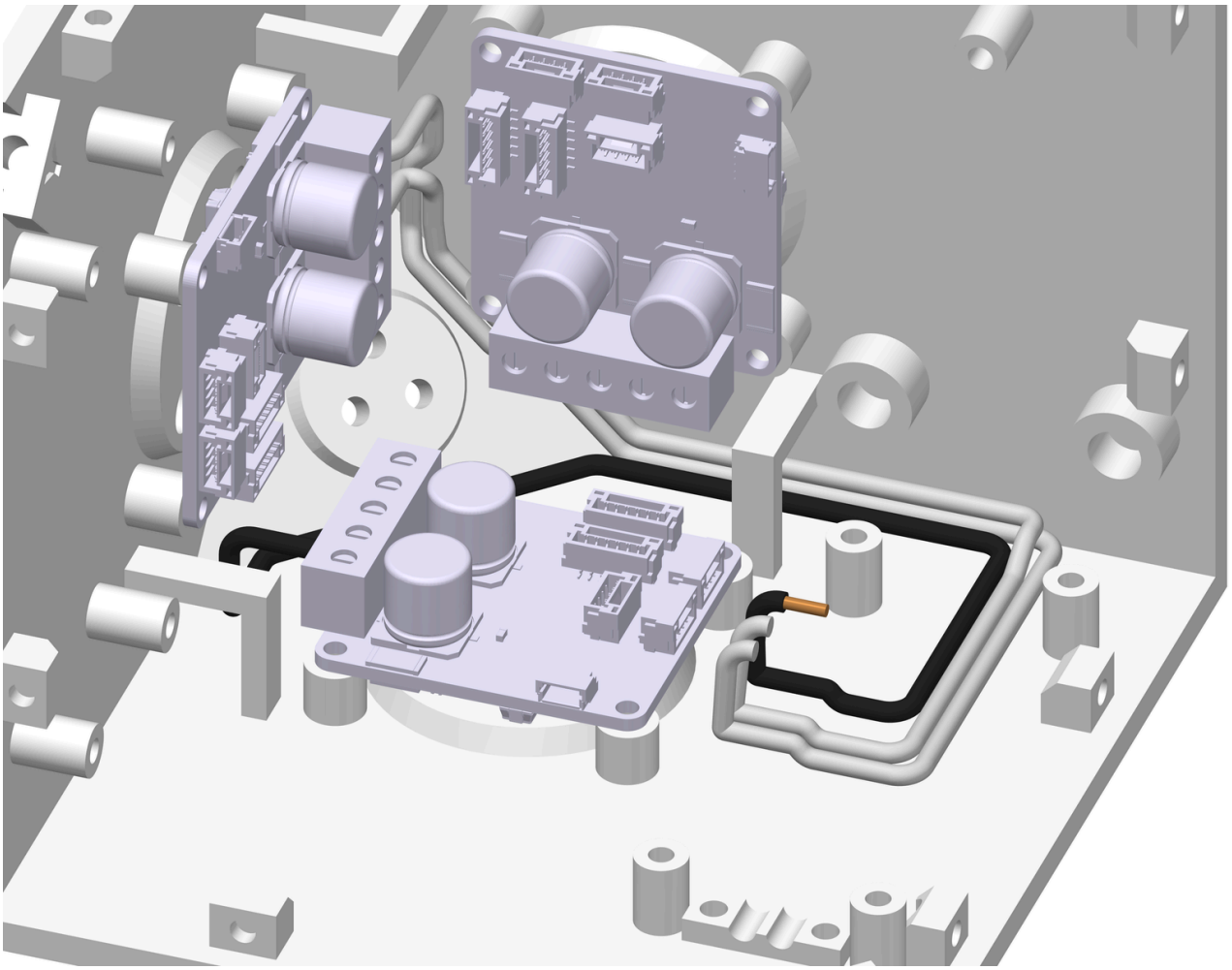
From driver Y (-) to board Z, z6 (M3-).
The lowest wire - it goes first.

M3+ · red



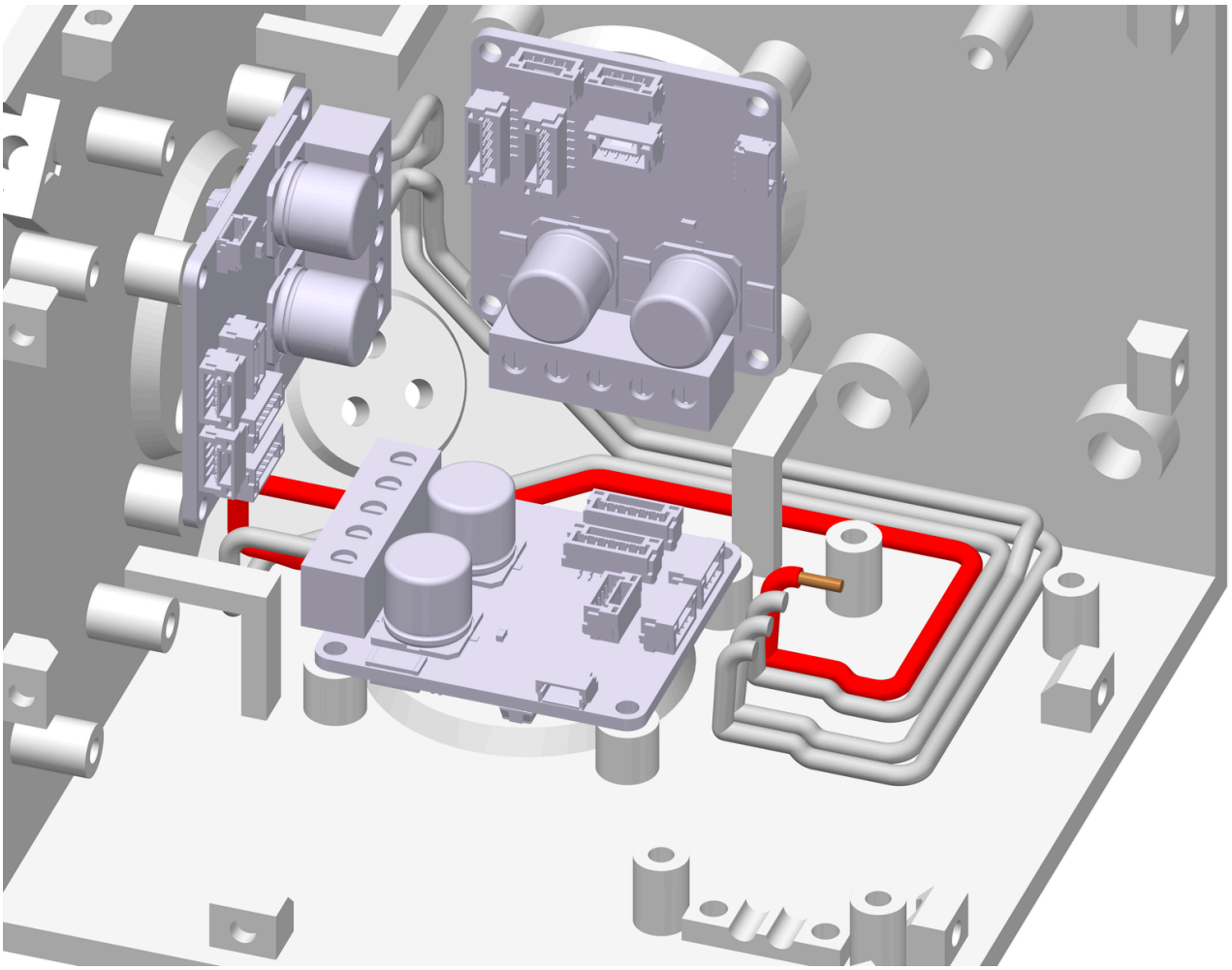
From driver Y (+) to board Z, z5 (M3+).
Along the edge it lies on M3-.

M2- · black



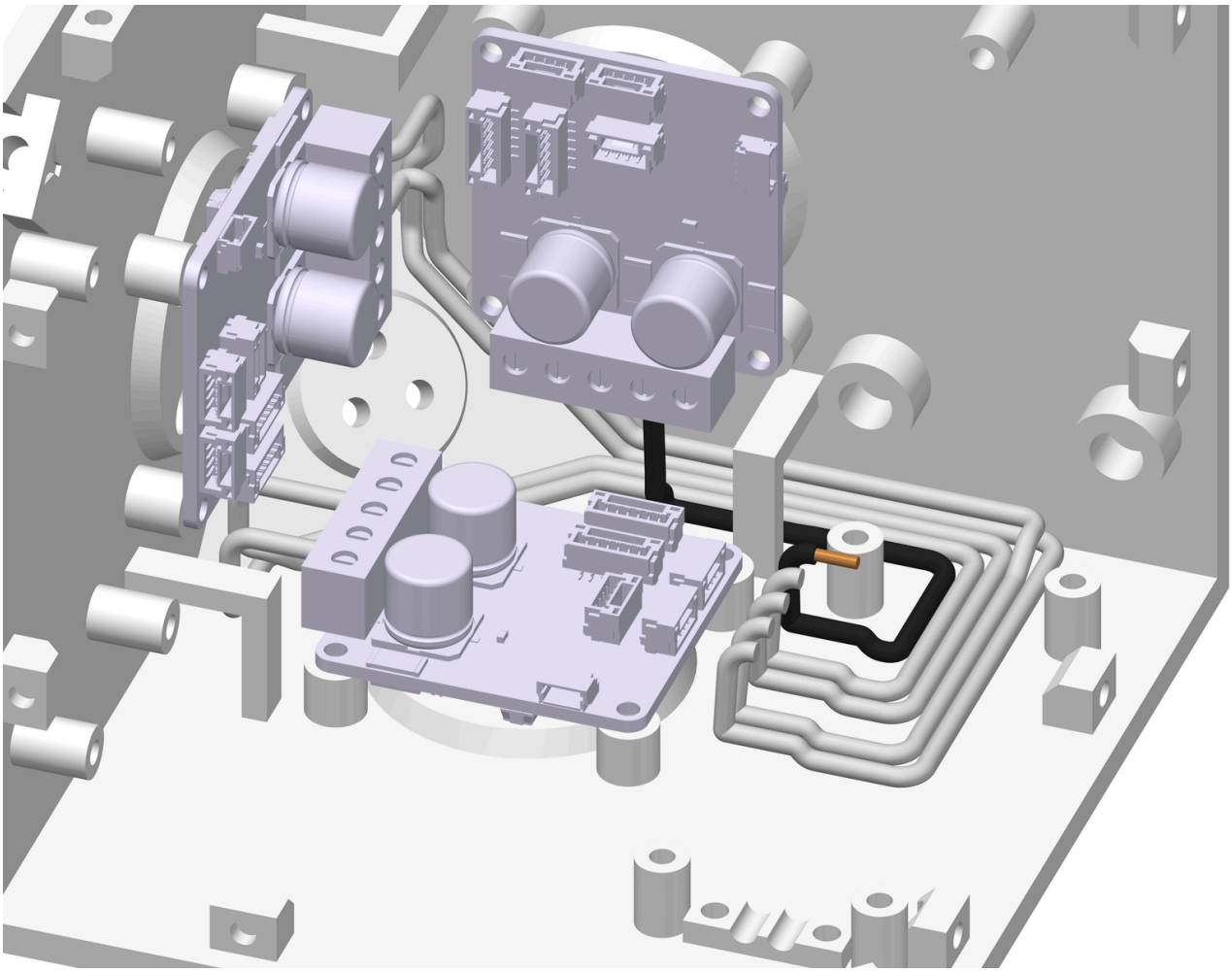
From driver Z (-) to board Z, z4 (M2-).
On the floor in its lane, through its pair's opening, the end up.

M2+ · red



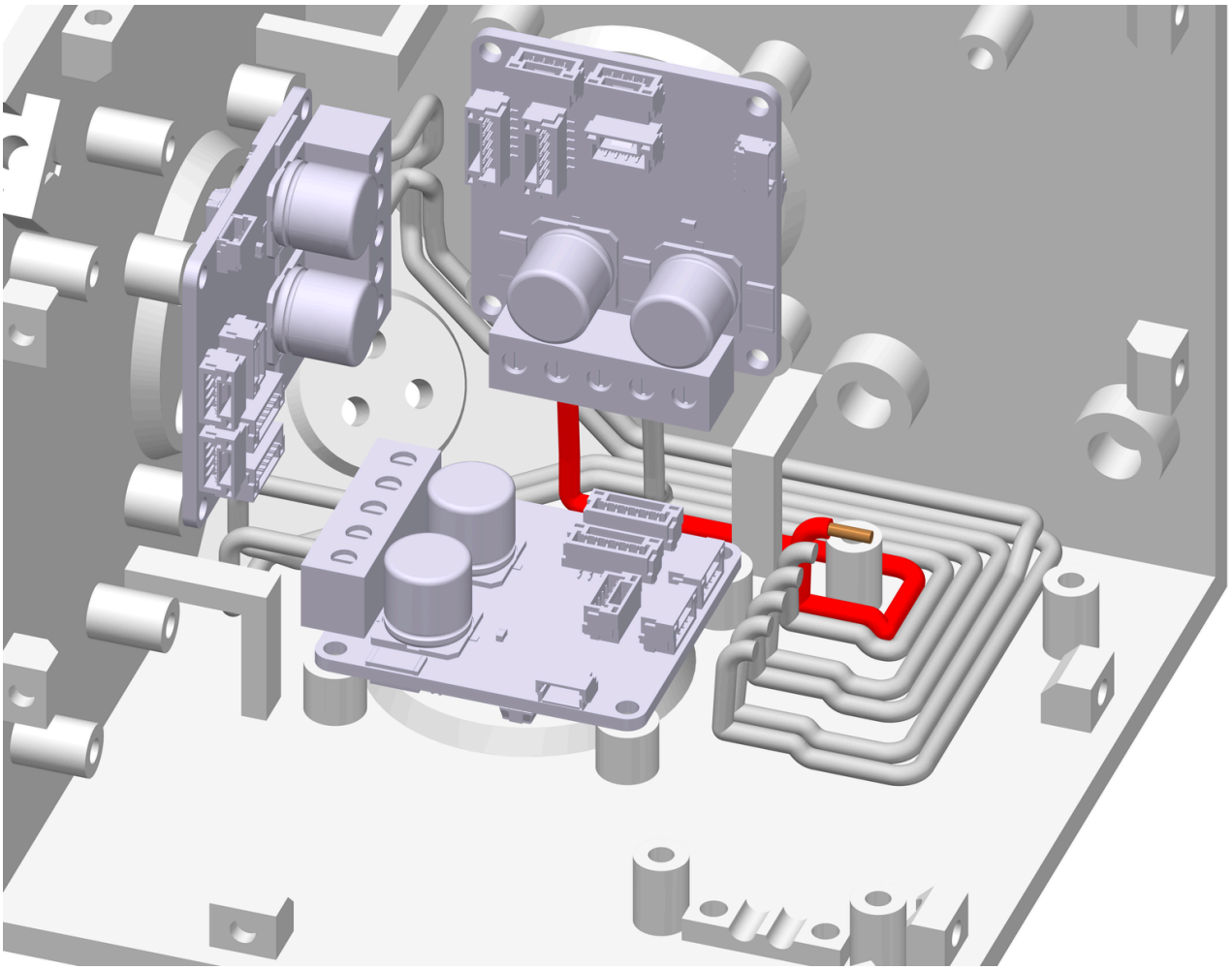
From driver Z (+) to board Z, z3 (M2+).
On the floor in its lane, through its pair's opening, the end up.

M1- · black



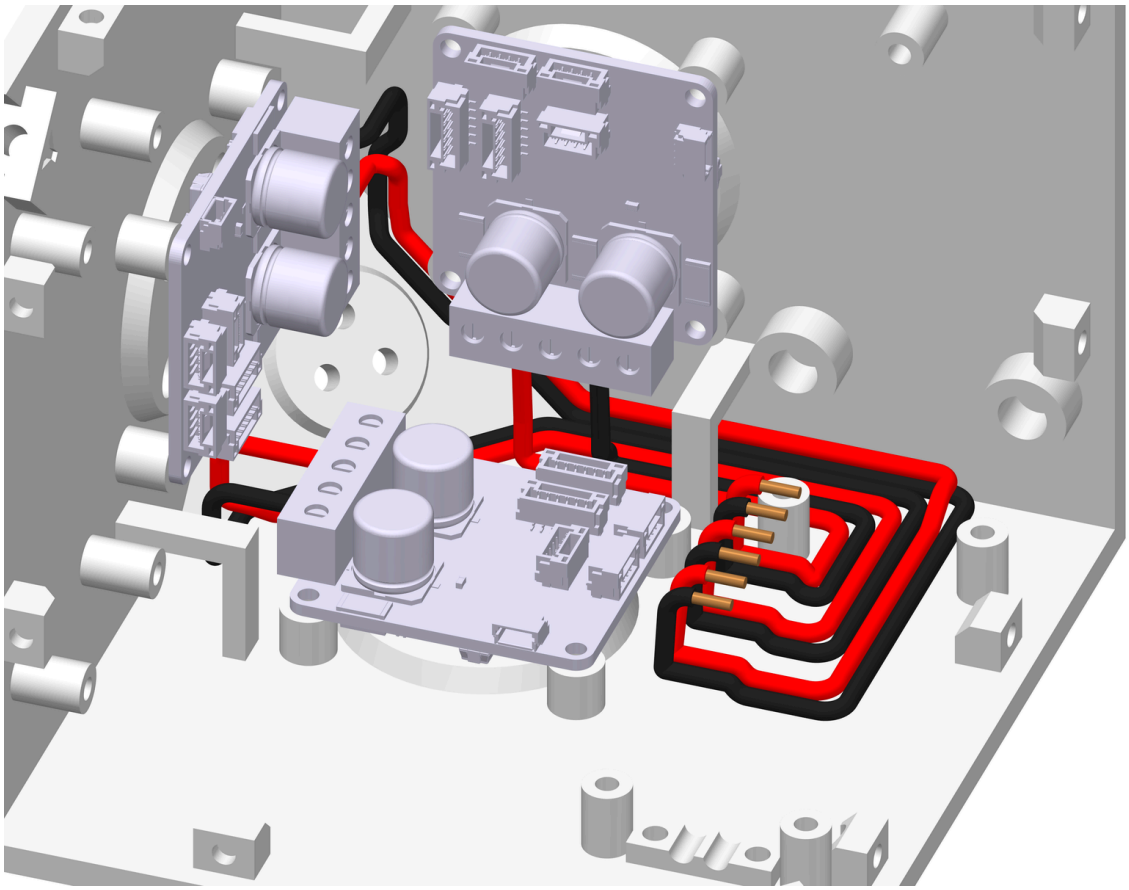
From driver X (-) to board Z, z2 (M1-).
On the floor in its lane, through its pair's opening, the end up.

M1+ · red



From driver X (+) to board Z, z1 (M1+).
On the floor in its lane, through its pair's opening, the end up.

Check · motor power



- 6 wires, red and black in turn, z1 to z6.
- No wire crosses another.
- Each wire on the floor where its arch of clamp_z will be, the ends standing up.

Cut list · battery supply

14 AWG silicone wire (Ø3.0). Length = route + 10 mm.

measure

cut

name flag

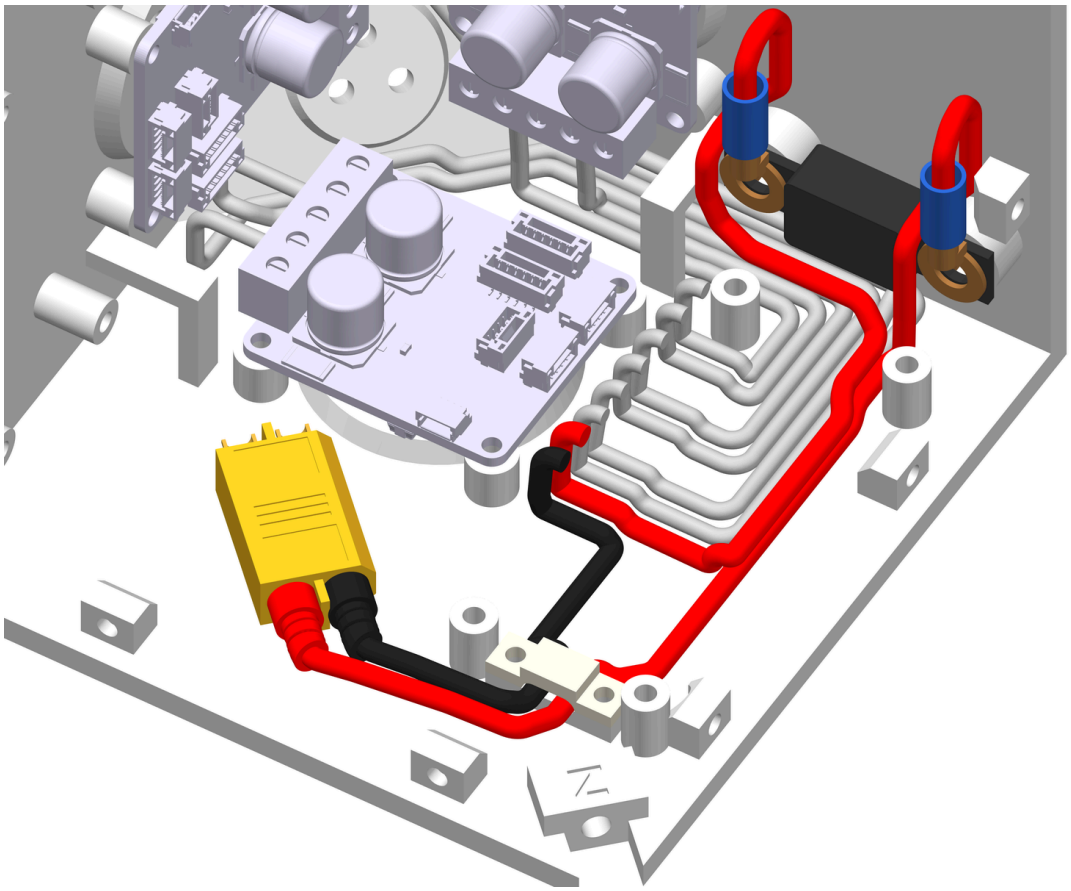
twist the ends

	color	from -> to	cut
1	red	Fuse -> battery connector	210 mm
2	black	z8 -> battery connector	100 mm
3	red	z7 -> fuse	190 mm

- Board end (z7, z8): strip 7 mm, twist the strands tight.
- Fuse end: a blue ring lug (DIN 46237, M5). Strip as long as its metal barrel, push the wire in, crimp with the blue jaw. No solder - the sleeve would melt.
- Battery connector end: soldered into the cup. Slide the heat-shrink on the wire BEFORE you solder.

2 · Battery supply

The battery current: battery connector -> fuse -> board Z (z7 +, z8 -). Three thick wires.

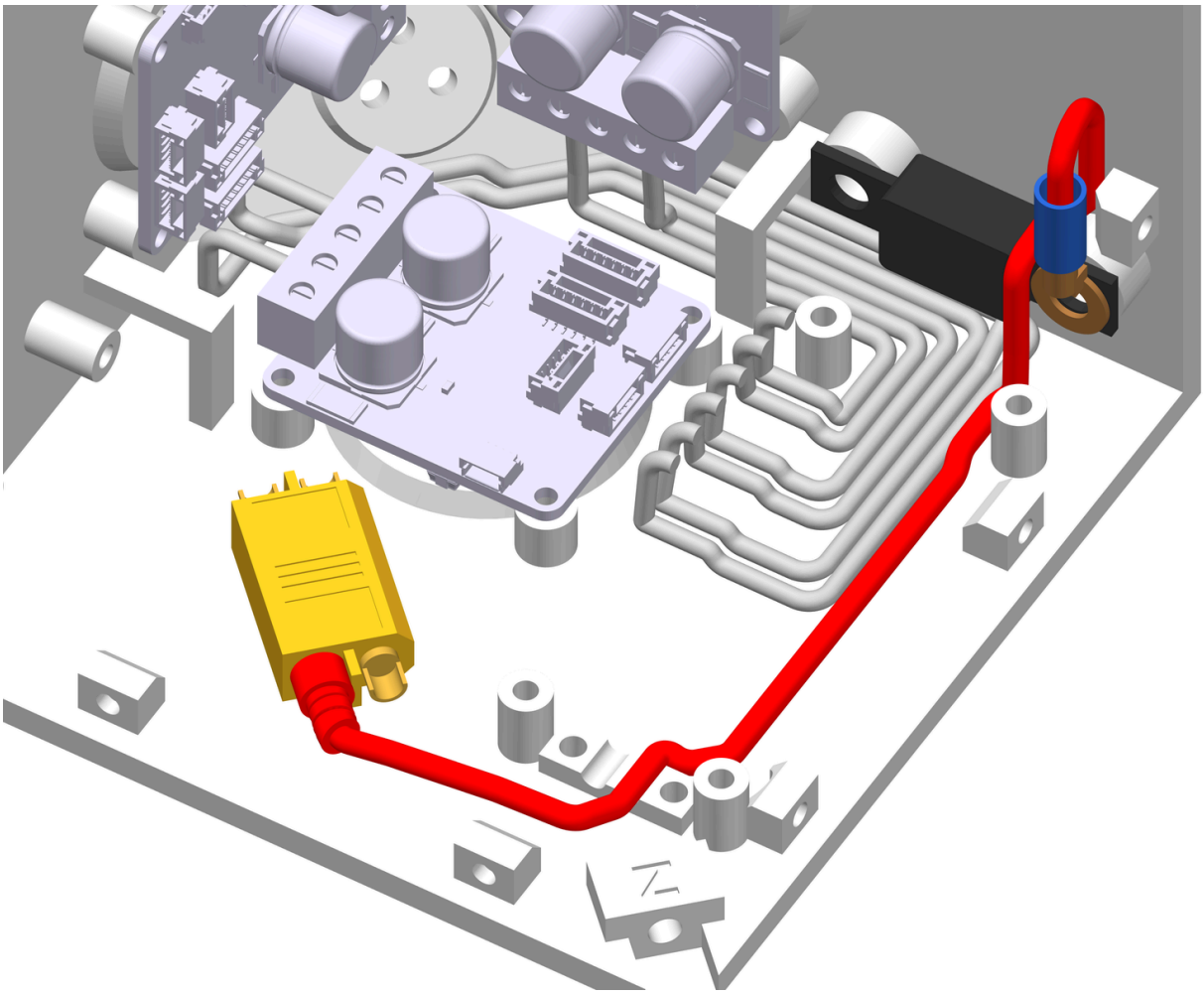


The two on the floor first, the one that lies on top last. At clamp_z the z7 (+) and z8 (-) wires each lie under their own arch.

Why?

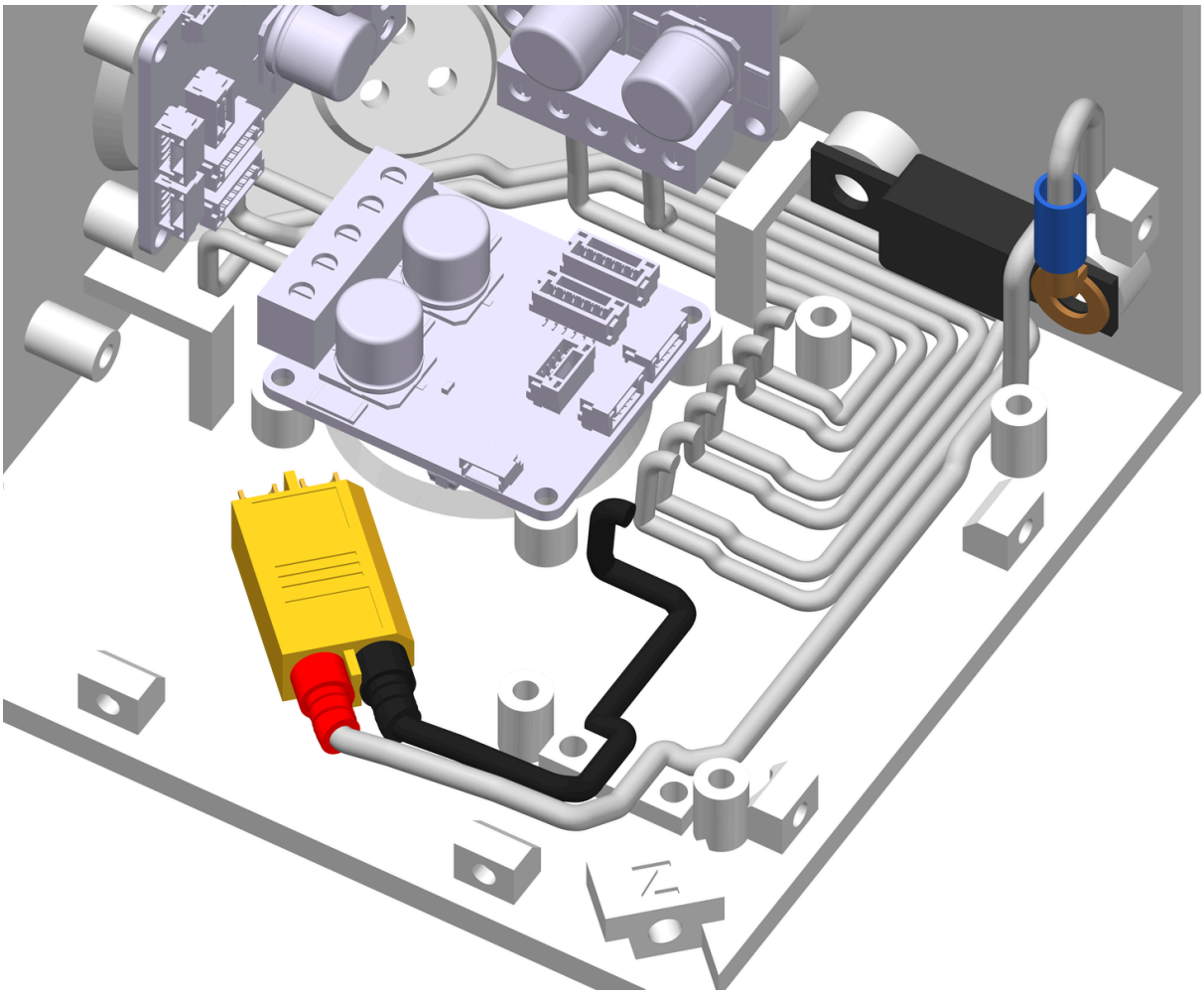
All the power of the cube comes through these three wires. The fuse is a thin strip of metal: if something shorts, it melts first and cuts the current - before a wire or the battery gets hot. It protects you and the cube.

Fuse -> connector · red



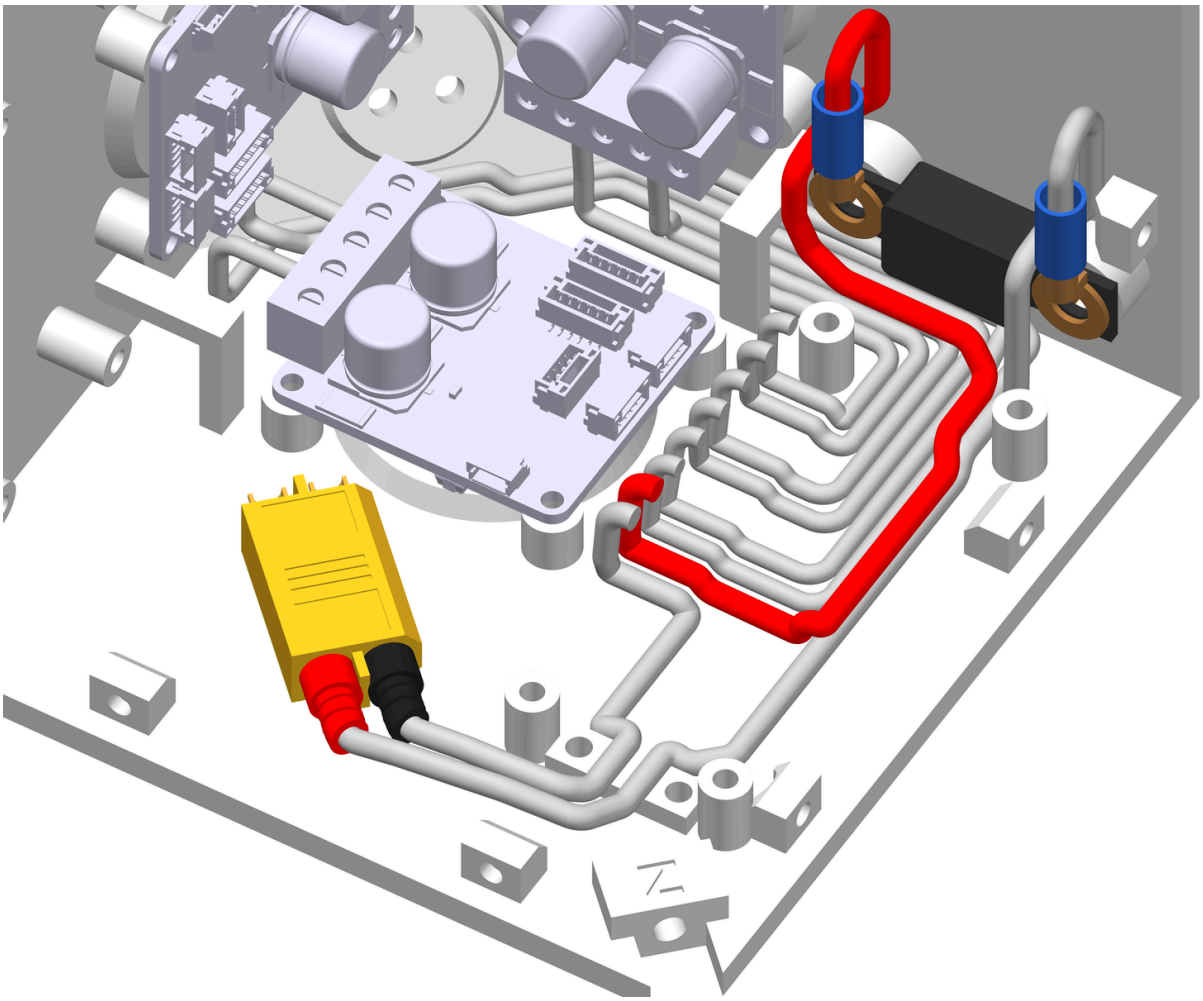
Far fuse screw -> under board Z beside M3- -> clamp -> connector +.
Blue ring lug at the fuse, crimped. At the connector: heat-shrink on first, then solder.

z8 -> connector • black



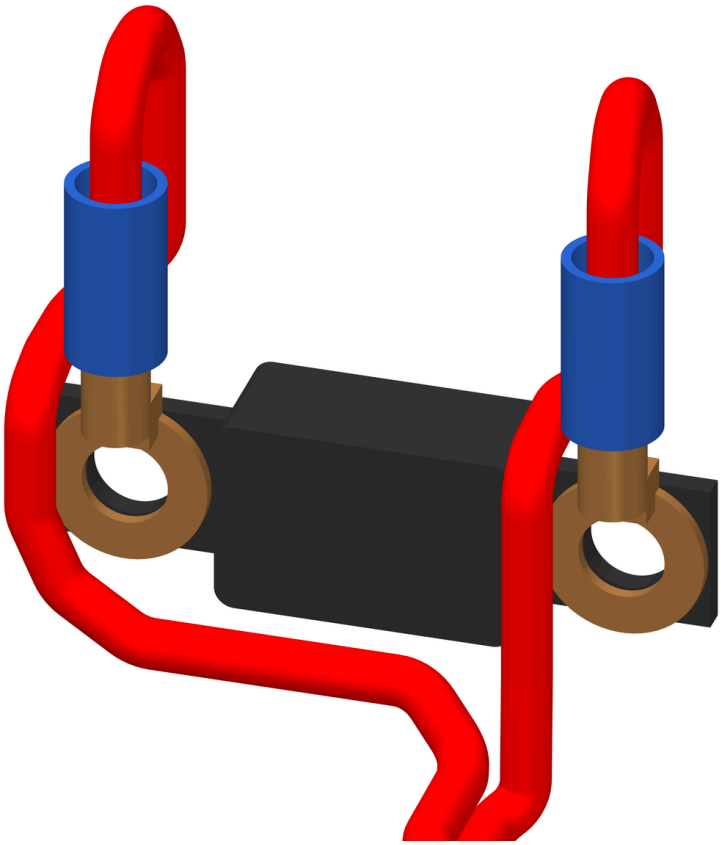
z8 (BAT-) -> on the floor -> clamp -> connector -.
Heat-shrink on first, then solder.

z7 -> fuse · red



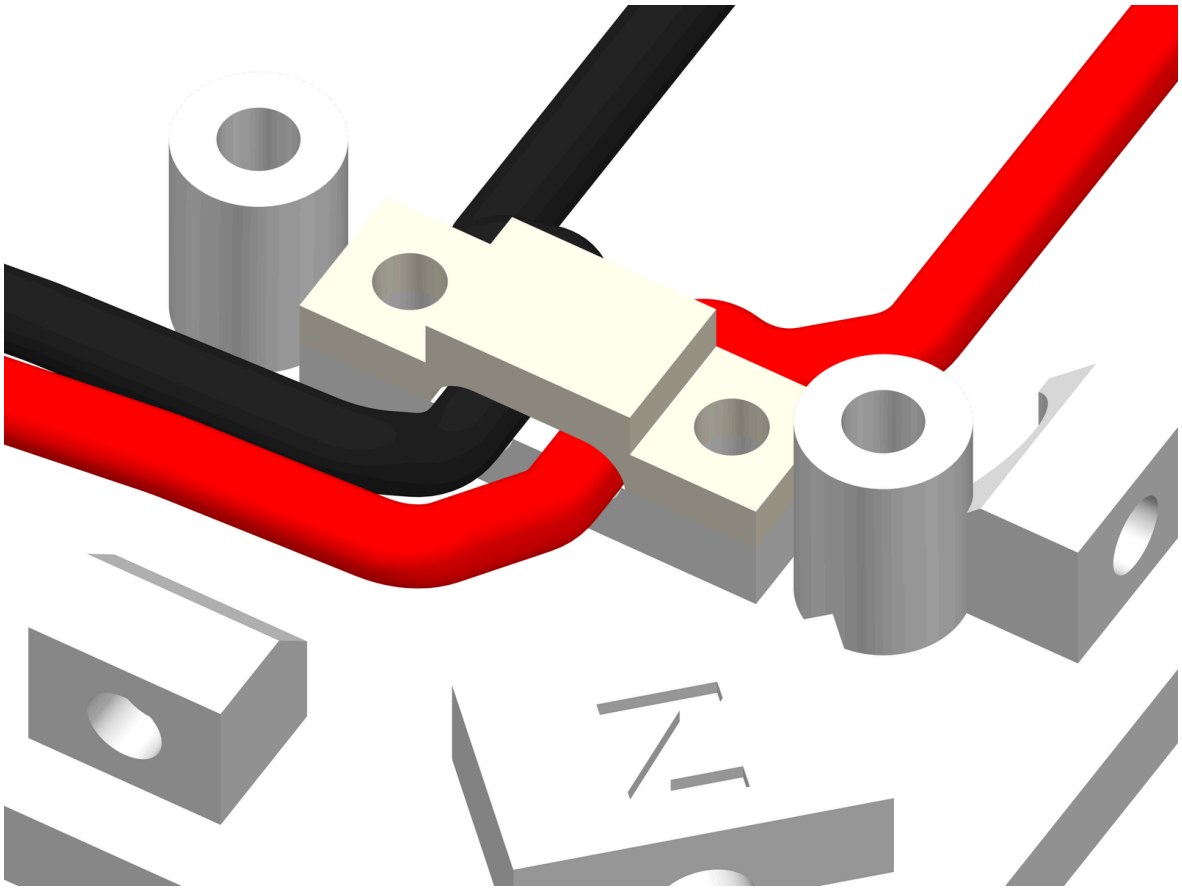
z7 (BAT+) -> up and on top of the first red -> near fuse screw.
Blue ring lug at the fuse, crimped.

Fuse: the two lugs



- Each lug on its M5 screw, straight UP: the blue sleeve on top.
- The wire comes up in front of the fuse, goes to the box wall over the lug, flat on the wall behind the sleeve, and over the top into it.
- Near screw: the wire from z7. Far screw: the wire to the connector.
- Tighten firmly - a loose lug gets hot.

The cable clamp



- The bar over the two thick wires, 2 screws M2.5 x 6.
- Firm, not crushing: the wires must not move when you pull them.

Teacher only: the battery side

The teacher's job

The other half of the connector goes on the battery's own leads. The battery is charged.

One lead at a time

Cut, heat-shrink on, solder, shrink - only then the other lead. Never both leads bare at the same time.

Red to +, black to -

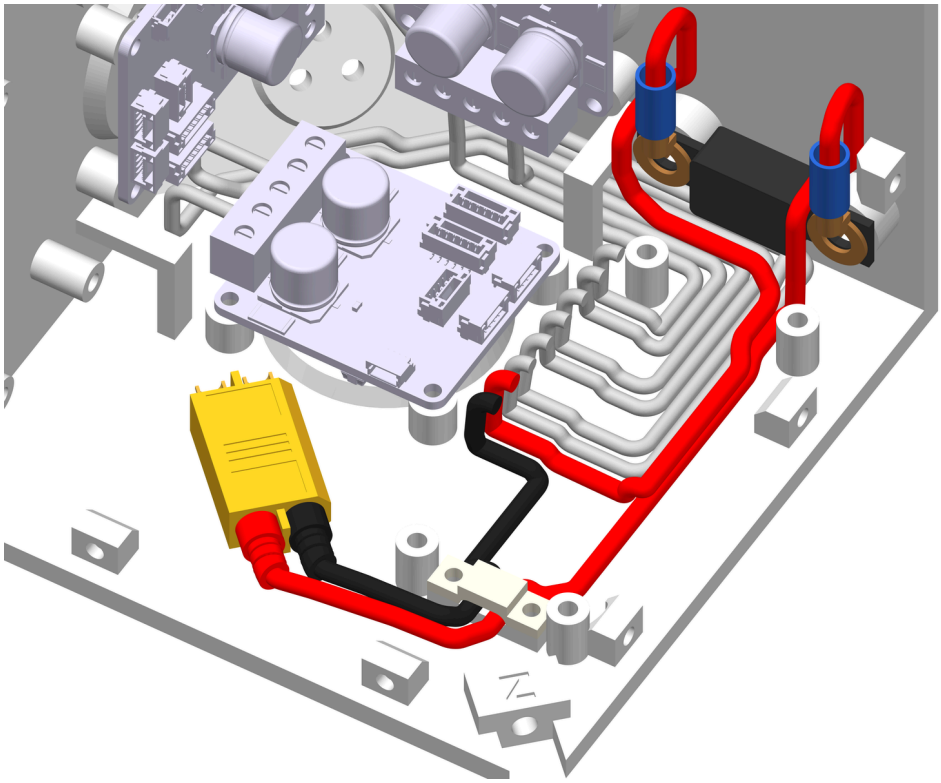
Check it twice.

Or ready-made

Buy the battery with this connector already on.

Check · battery supply

Multimeter on beep. NO battery.



- z7 -> near fuse lug: beep.
- Far fuse lug -> connector +: beep.
- z8 -> connector -: beep.
- Connector + -> connector -: NO beep. A beep here is a short circuit - find it before anything else.

Cut list · board power

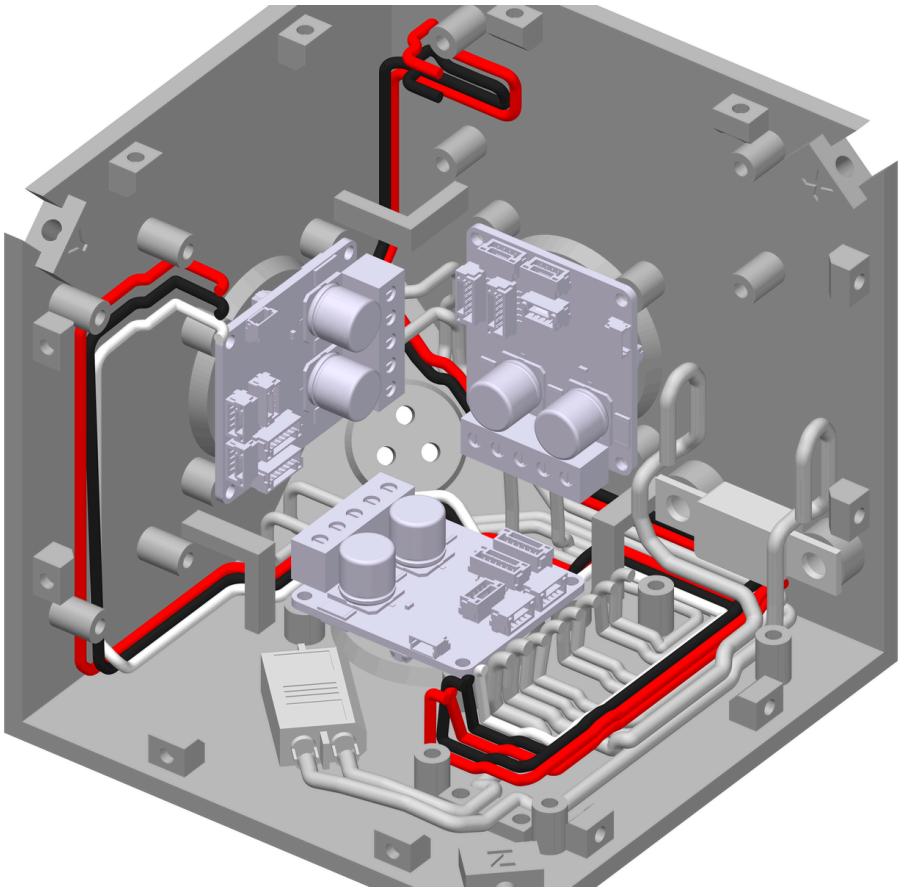
0.75 mm2 silicone wire (Ø2.1). Length = route + 10 mm.

	color	from -> to	cut
1	white	z9 SENSE -> y3 SENSE	350 mm
2	black	z10 GND -> y2 GND	365 mm
3	red	z11 +5V -> y1 +5V	385 mm
4	black	z10 GND -> x2 GND	330 mm
5	red	z11 +5V -> x1 +5V	345 mm

- Strip 7 mm on each end, twist the strands tight.
- At board Z, GND and +5V take TWO wires each (one to Y, one to X): twist the two together and put them in as one, under the clamp plate.

3 · Board power

Board Z makes the 5 V. Thin wires carry it to board Y and to board X. They lie ON the motor power wires.



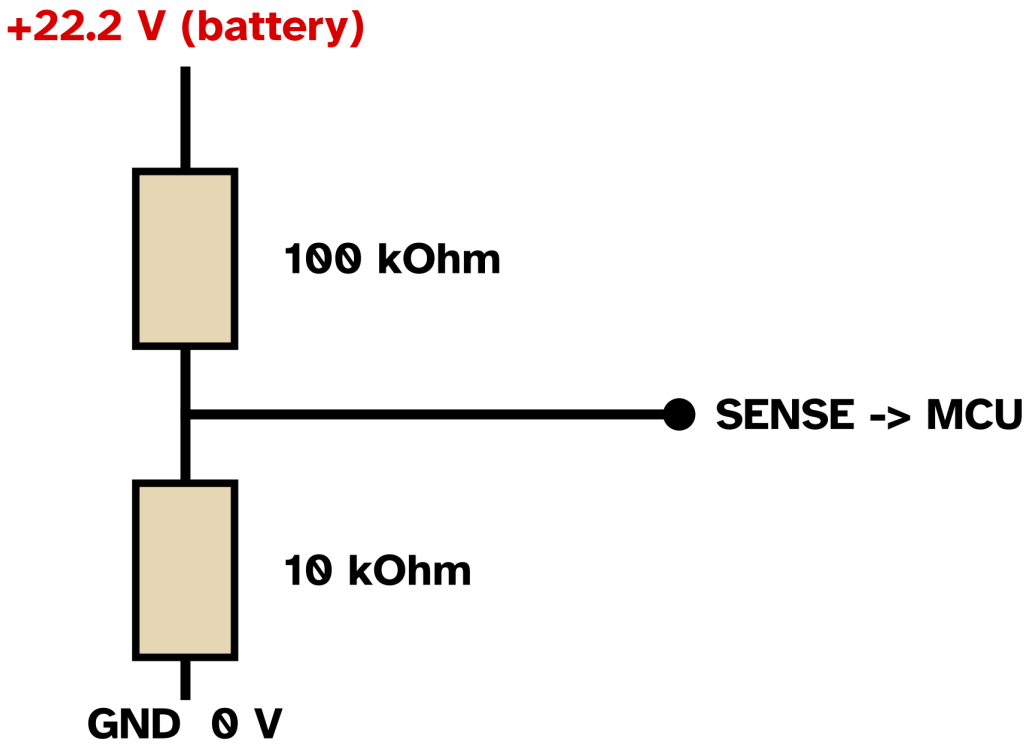
Z>Y (GND + +5V) and Z>X (GND + +5V): every wire under its own arch of the clamps.

Why?

The small chips on the boards (the brain, the radio, the IMU) live on 5 V or less. 22 V from the battery would burn them. So board Z turns 22 V into 5 V, and these wires bring it to the other boards: +5V (red) and GND (black).

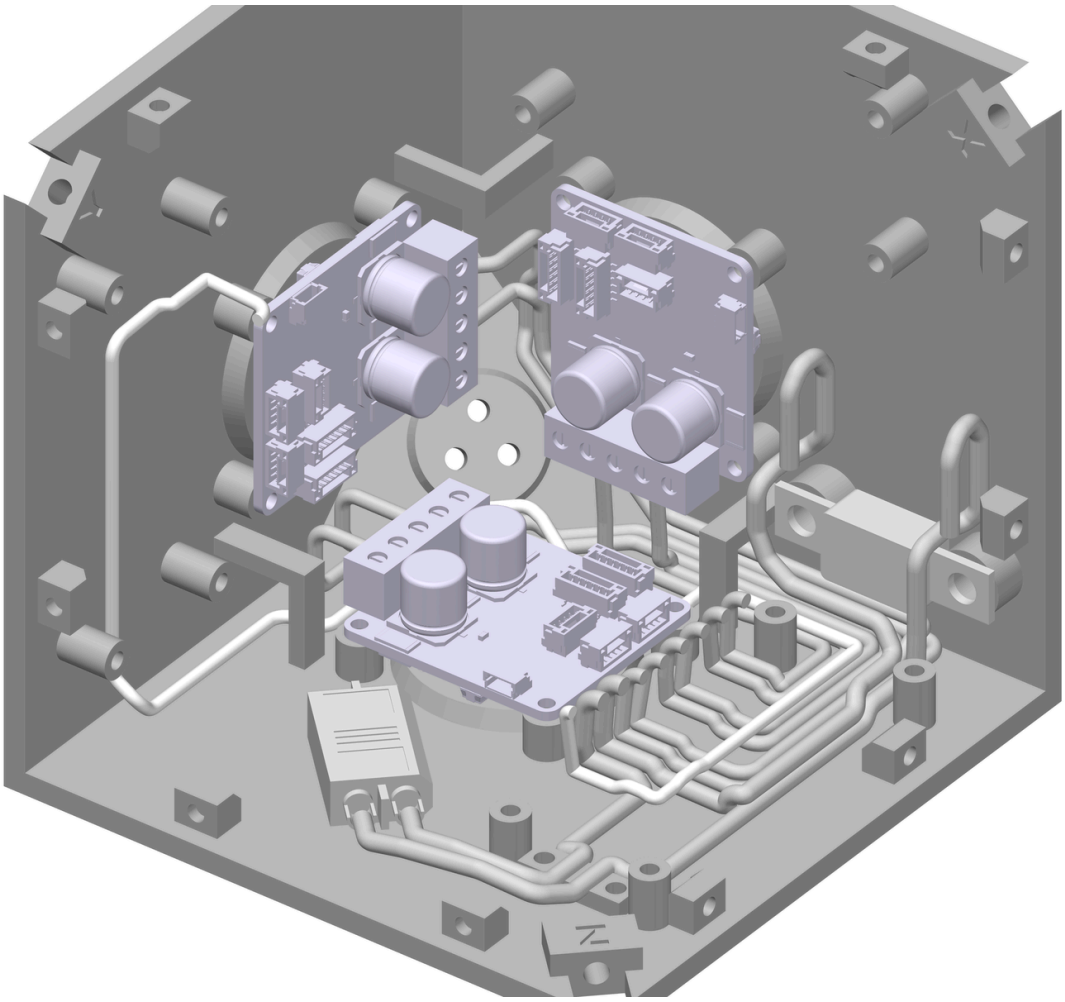
The white wire: SENSE

How does the cube know how full its battery is?



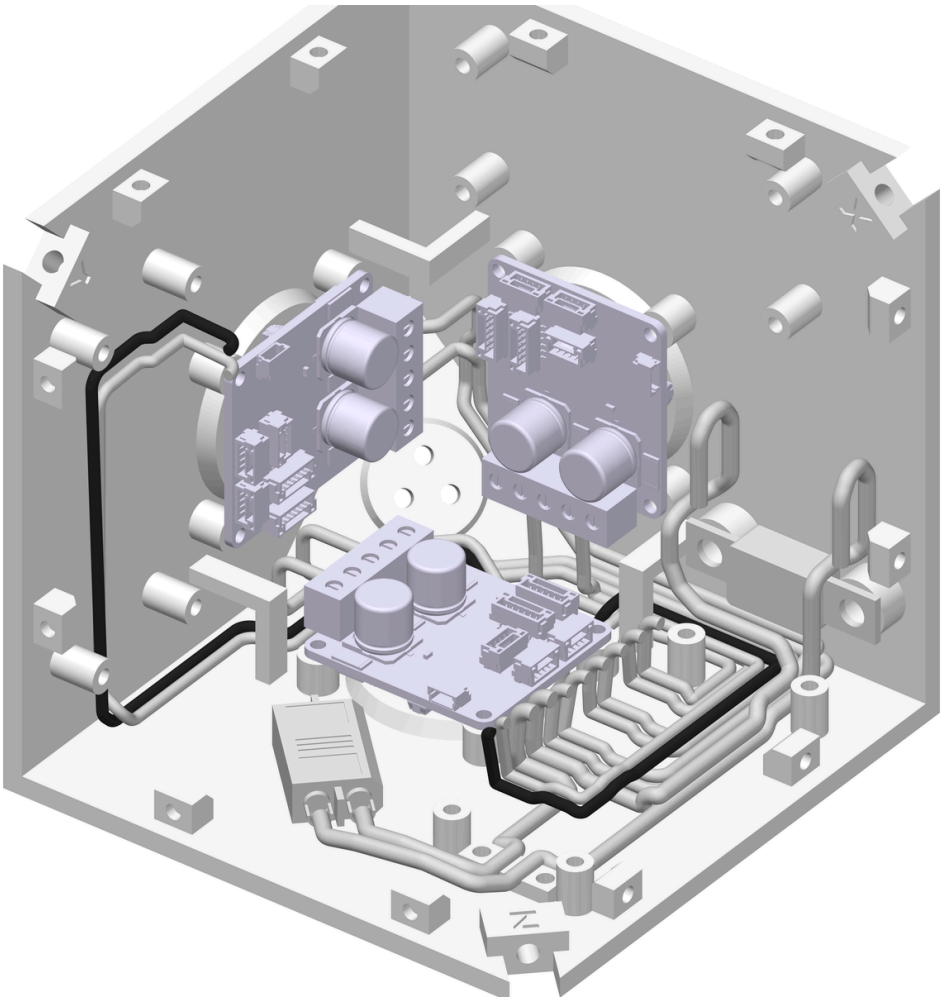
- The brain (MCU) can measure a voltage - but only up to 3.3 V.
- On board Z two resistors, 100 kOhm and 10 kOhm, share the battery voltage: the white wire gets 1/11 of it.
- Full battery 25.2 V -> 2.3 V on SENSE. Low battery 21 V -> 1.9 V.
- The MCU reads it and knows when the battery needs charging - like the battery bar on a laptop.

SENSE -> board Y · white



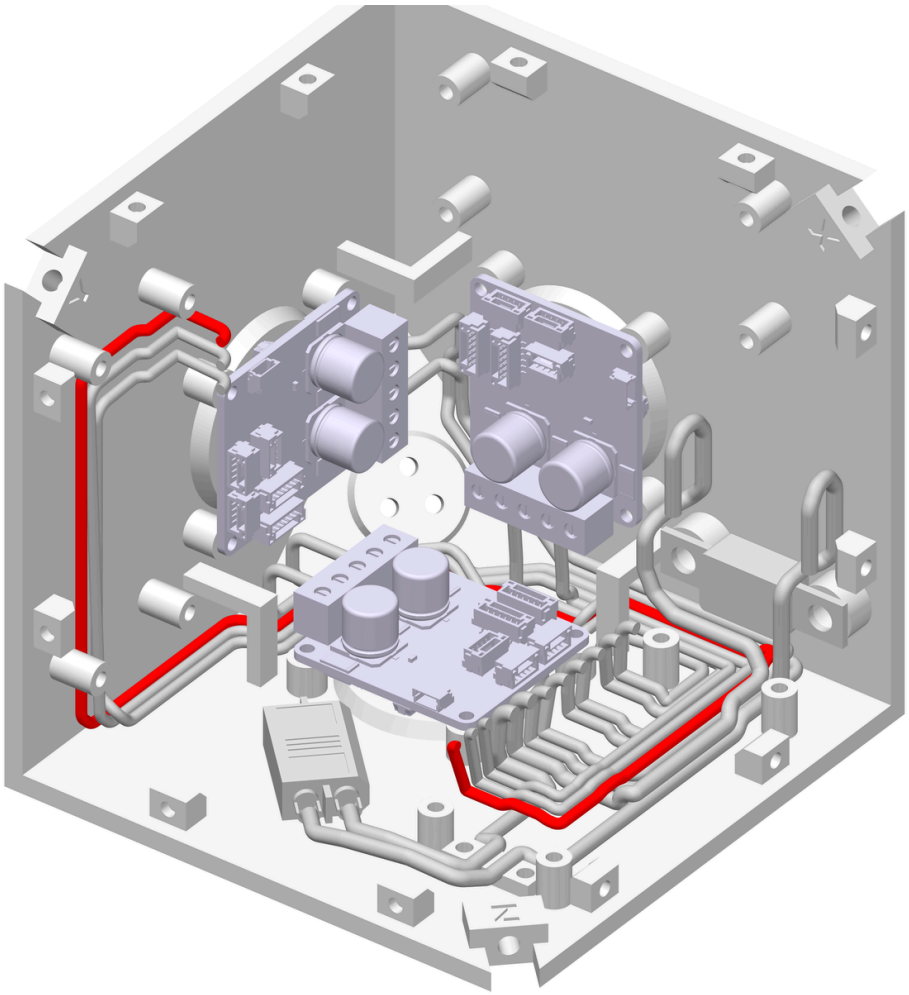
Board Z SENSE (z9) -> board Y SENSE (y3).
On top of the thick wires, along the same edges.

GND -> board Y · black



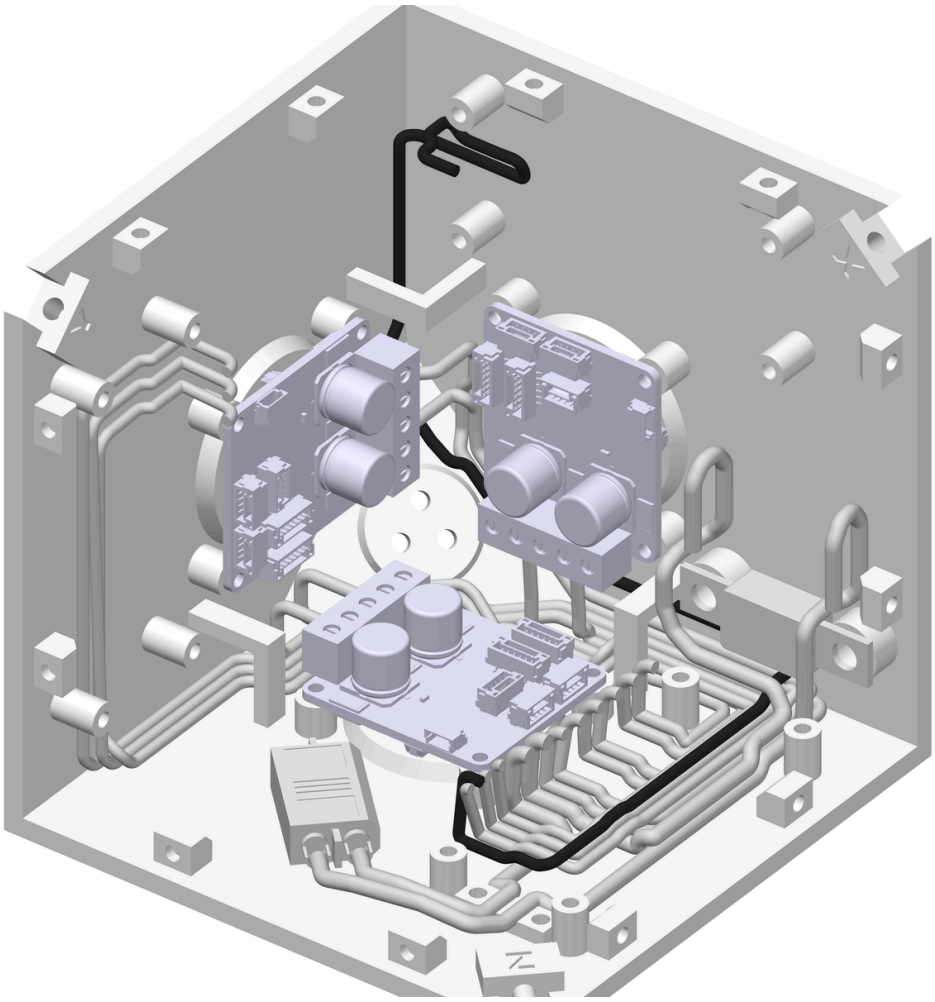
Board Z GND (z10) -> board Y GND (y2).
Its own arch in the clamps, on top of the thick wires.

+5V -> board Y · red



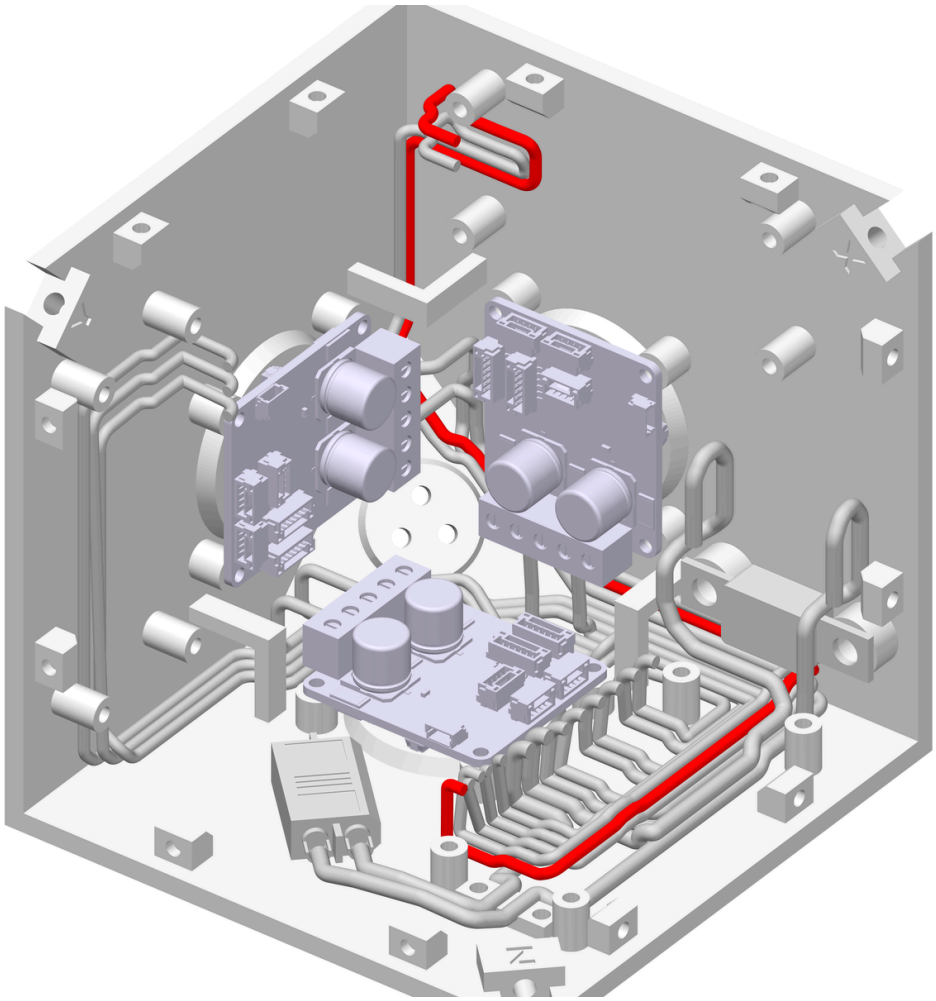
Board Z +5V (z11) -> board Y +5V (y1).
Its own arch in the clamps, on top of the thick wires.

GND -> board X · black



Board Z GND (z10) -> board X GND (x2).
Its own arch in the clamps. At board X: in under the narrow side with the other wires, out through clamp_x_1.

+5V -> board X · red



Board Z +5V (z11) -> board X +5V (x1).
Its own arch in the clamps. At board X: in under the narrow side with the other wires, out through clamp_x_1.

Cut list · signals

Links: 22 AWG solid wire, length = route + 10 mm. Cables: ready-made 4-wire cables, at least this long.

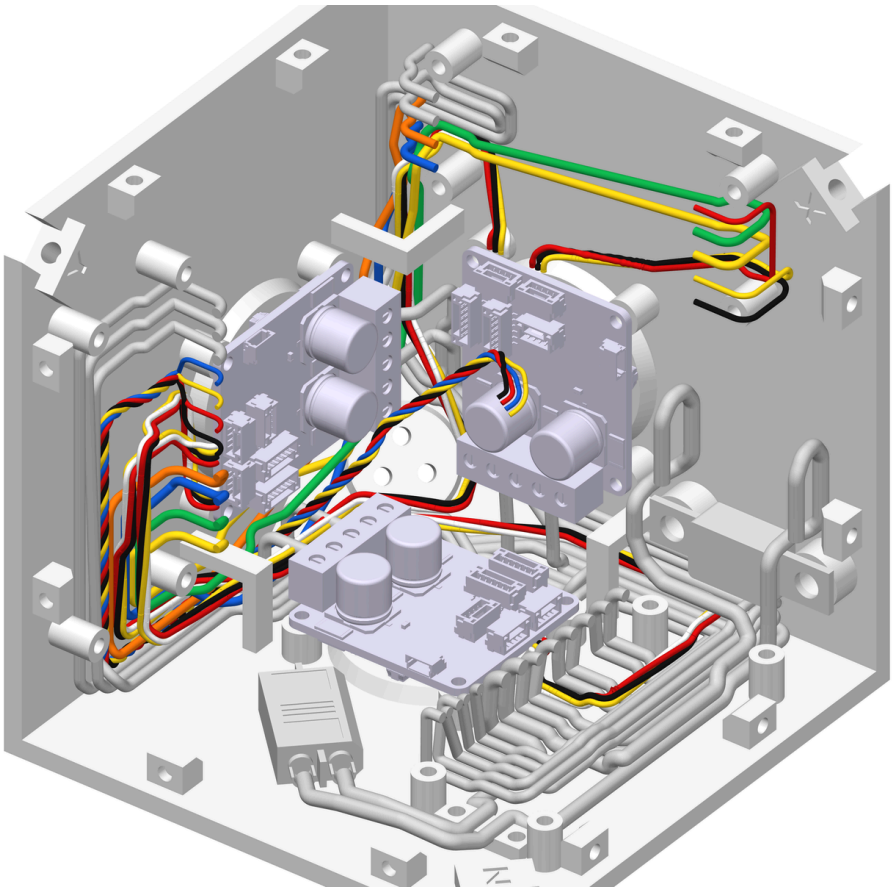
measure cut name flag twist the ends

wire or cable	cut
Link CRX2 · yellow	265 mm
Link CTX2 · green	275 mm
Link BRX · orange	235 mm
CAN 3 · driver Y -> driver Z	min 260 mm
Link BTX · blue	220 mm
I2C · IMU -> board Y	min 200 mm
CAN 2 · driver X -> driver Y	min 280 mm
CAN 1 · driver X -> board X	min 60 mm

- Links: solid wire - strip 7 mm, straight into the terminal.
- I2C and CAN 1: cut the cable where its wires part. Each thin wire is soldered to 3 cm of solid 22 AWG wire, heat-shrink over the joint - that end goes into the terminal.
- CAN 1 has 4 wires, board X needs 3: the WHITE one (NC) comes out of the plug - lift the plug's latch with a needle, pull the white wire out with its pin, peel it off the cable. The termination plug: the same for its red and white.

4 · Signals

The boards talk to each other (links), to the IMU (I2C) and to the motor drivers (CAN). Thin wires, last on the pile.

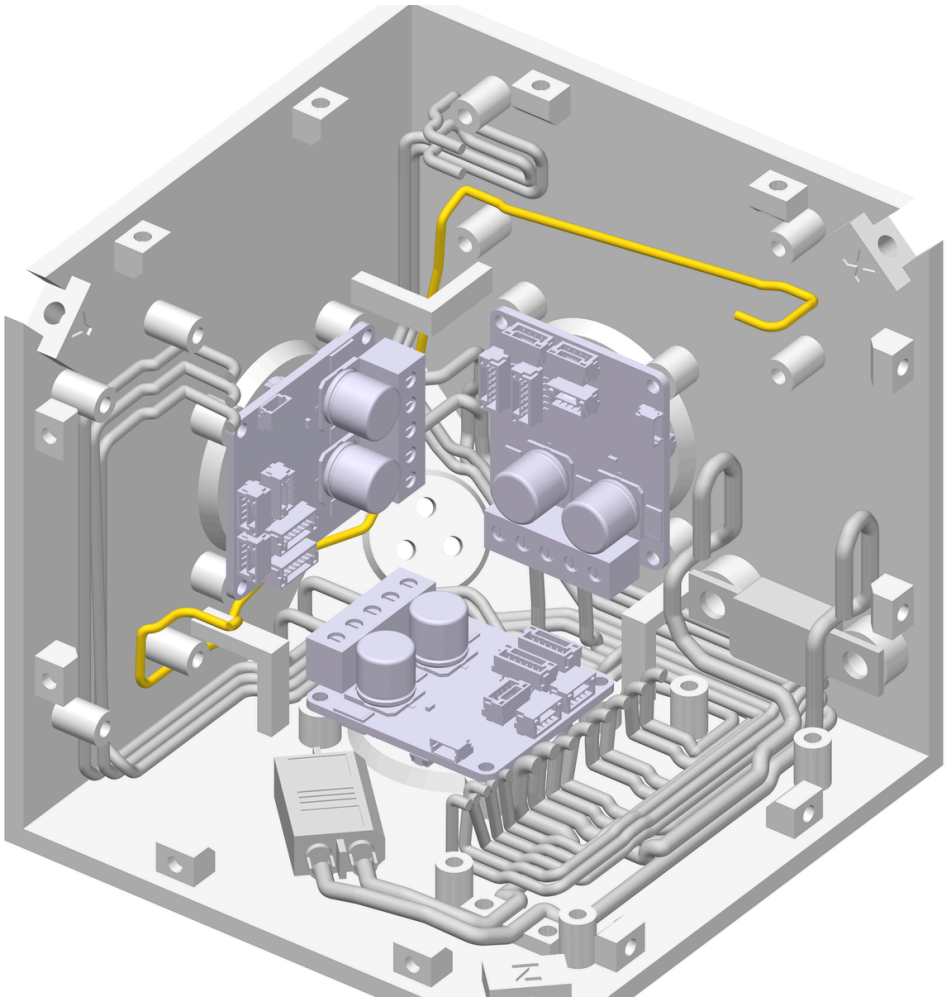


Order: under board Y they lie flat side by side, from the board edge inward.

Why?

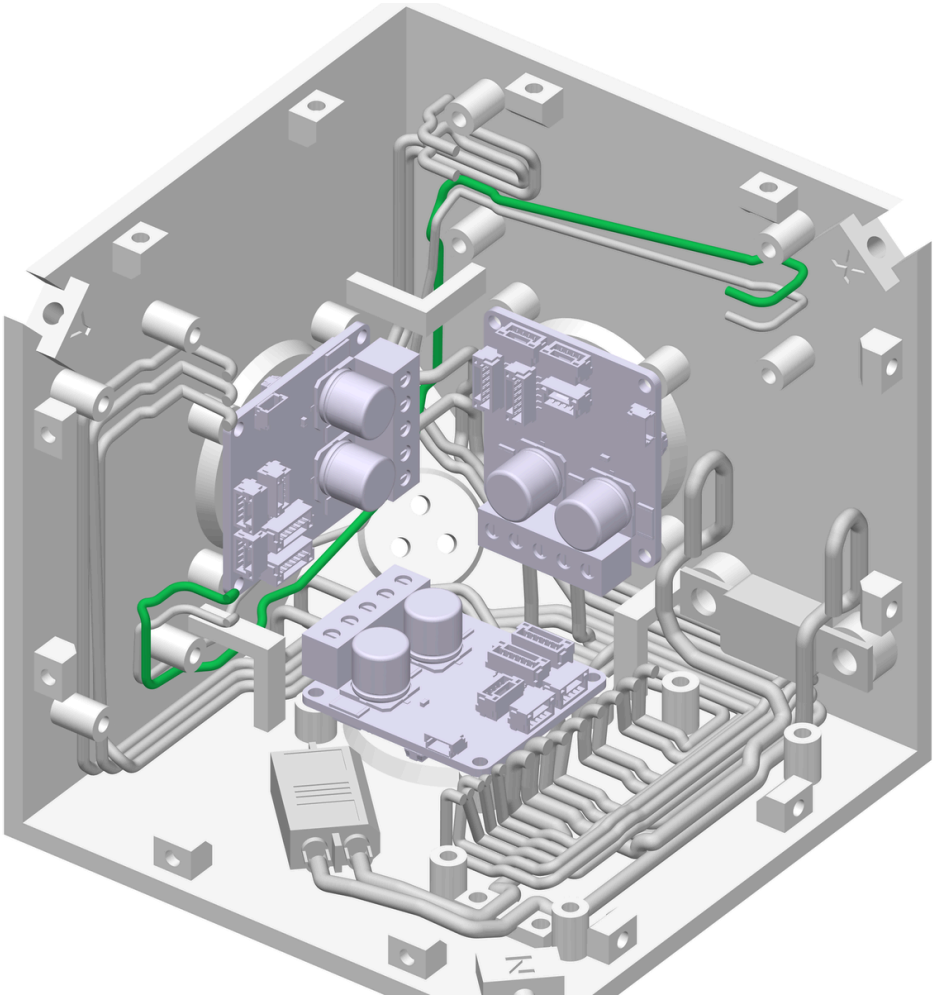
Links: board Y (the brain) and board X (radio + CAN) send each other messages. CAN: the brain tells each motor driver how hard to spin. I2C: the IMU tells the brain how the cube is tilted - many times a second. That is how Cubi 2 knows what to do.

Link CRX2 · yellow



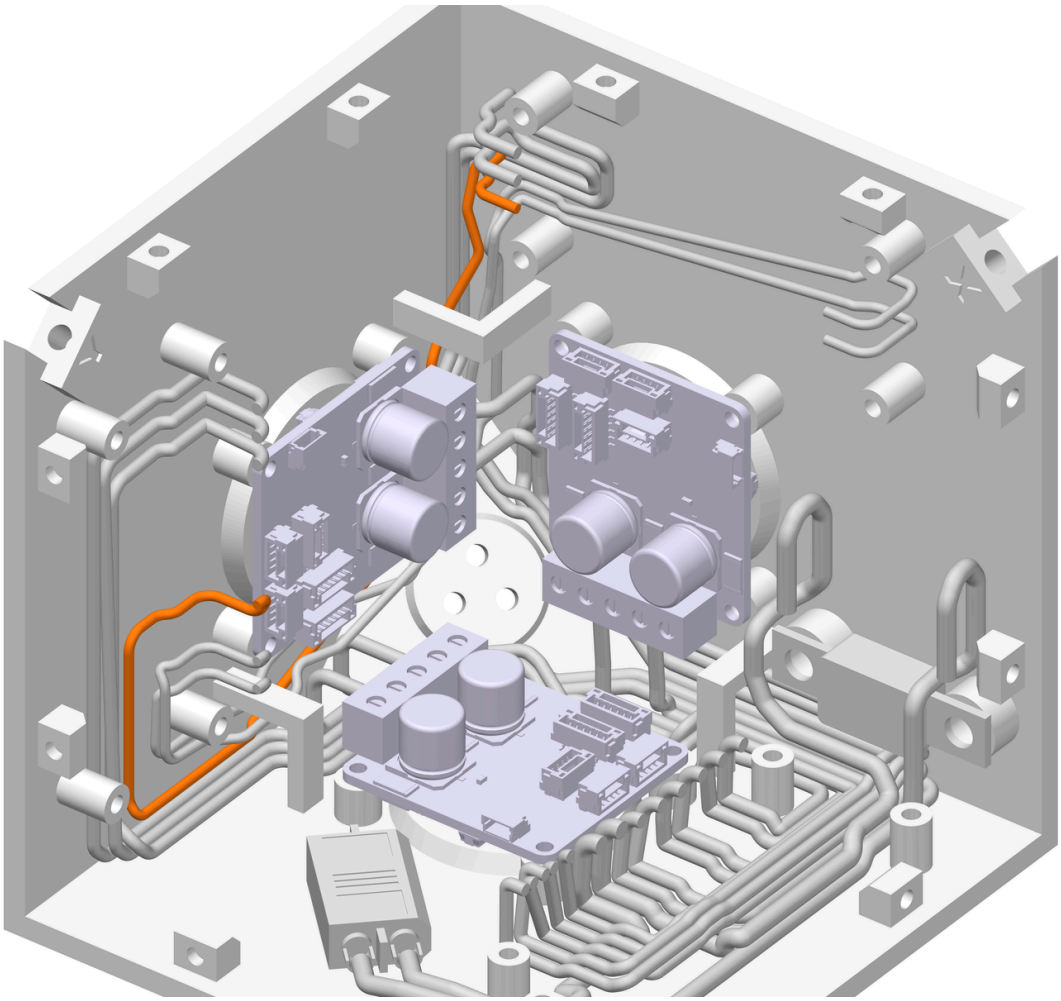
Board Y CRX2 (y11) -> board X CRX2 (x7).
In the clamps it has its own arch, beside CTX2.

Link CTX2 · green



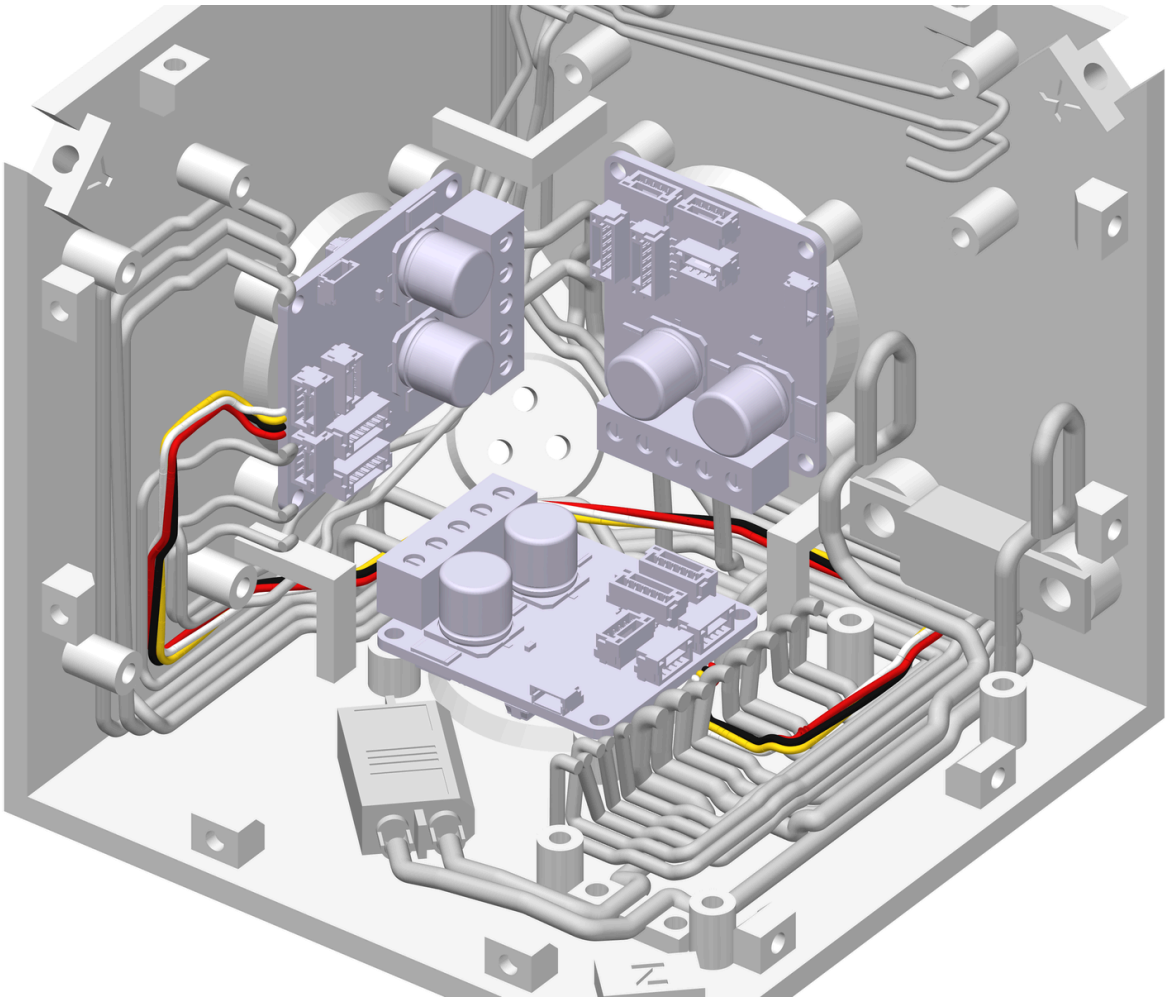
Board Y CTX2 (y10) -> board X CTX2 (x6).
In the clamps it has its own arch, beside CRX2.

Link BRX · orange



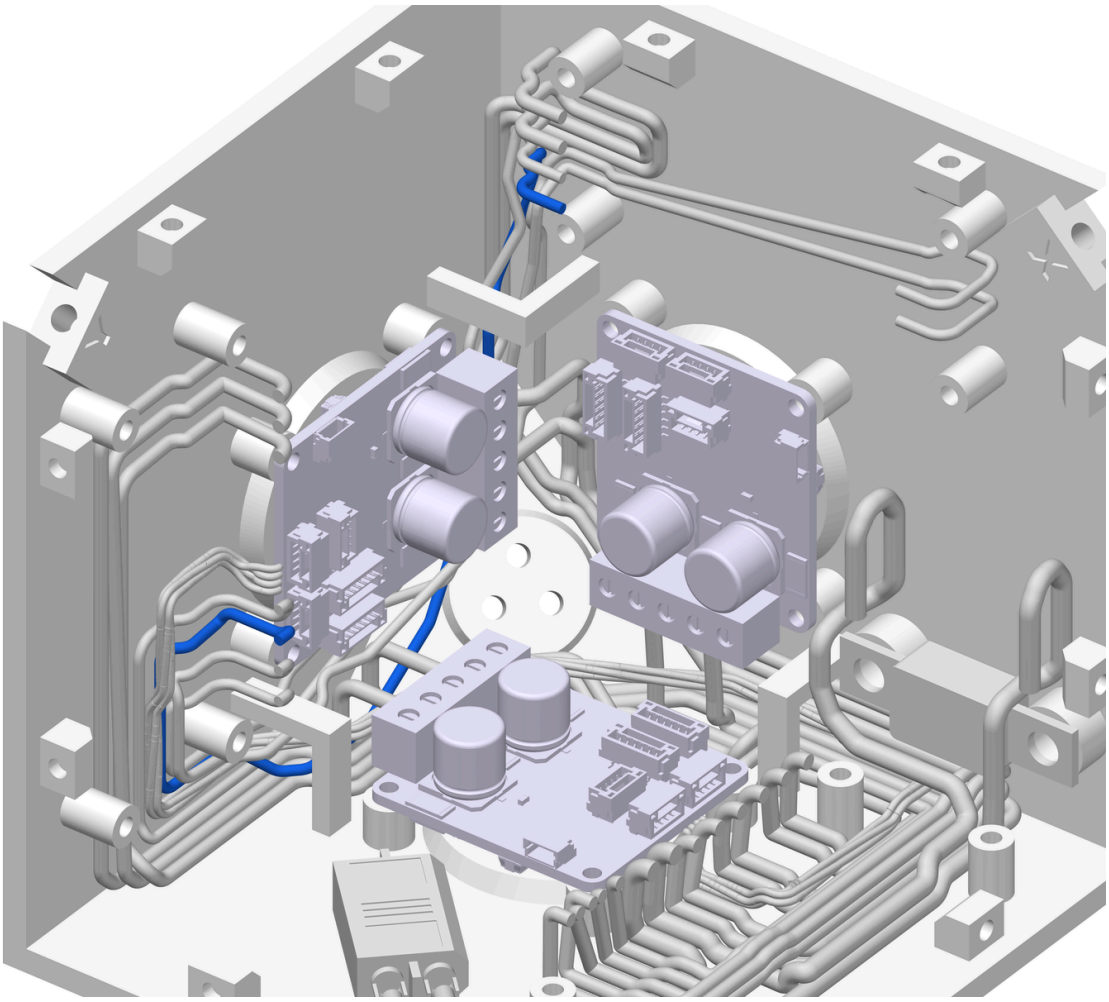
Board Y BRX (y8) -> board X BRX (x3).
In the clamps it has its own arch, beside BTX.

CAN 3 · driver Y -> driver Z



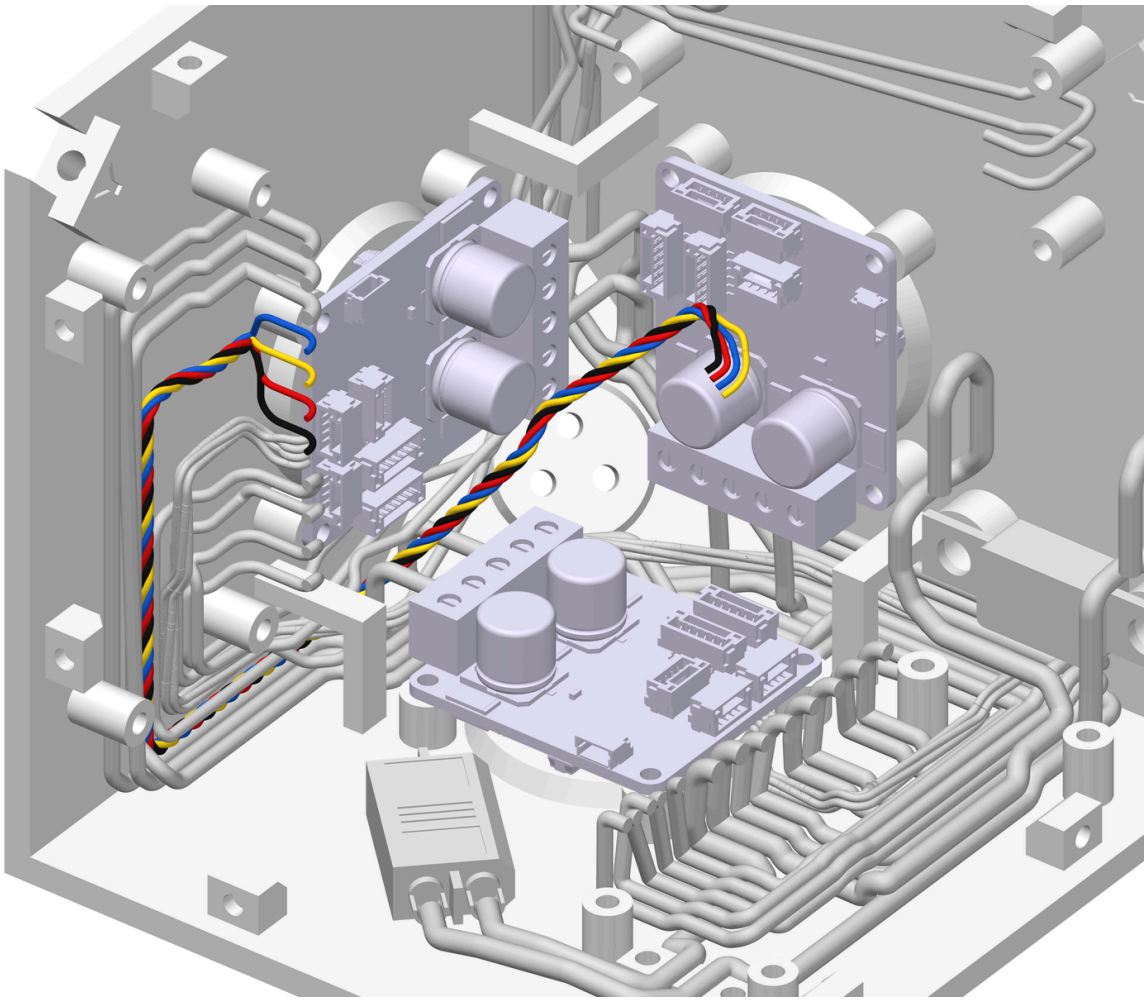
Driver Y (port 14) -> driver Z (port 14).
Plugs stay loose for now.

Link BTX · blue



Board Y BTX (y9) -> board X BTX (x4).
In the clamps it has its own arch, beside BRX.

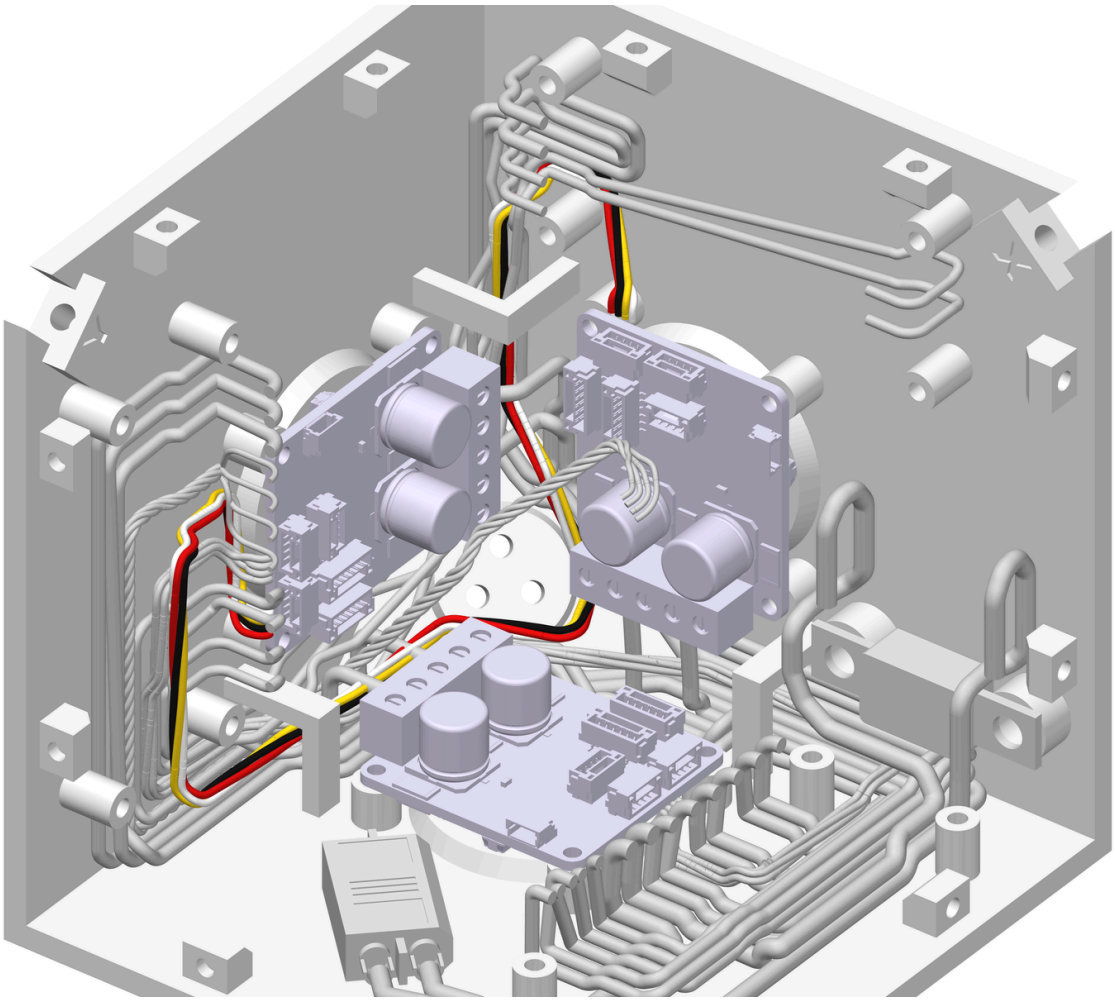
I2C · IMU -> board Y



IMU -> board Y: the cable round through clamp_y, then its 4 wires up: SDA y4, SCL y5, 3V3 y6, GND y7.

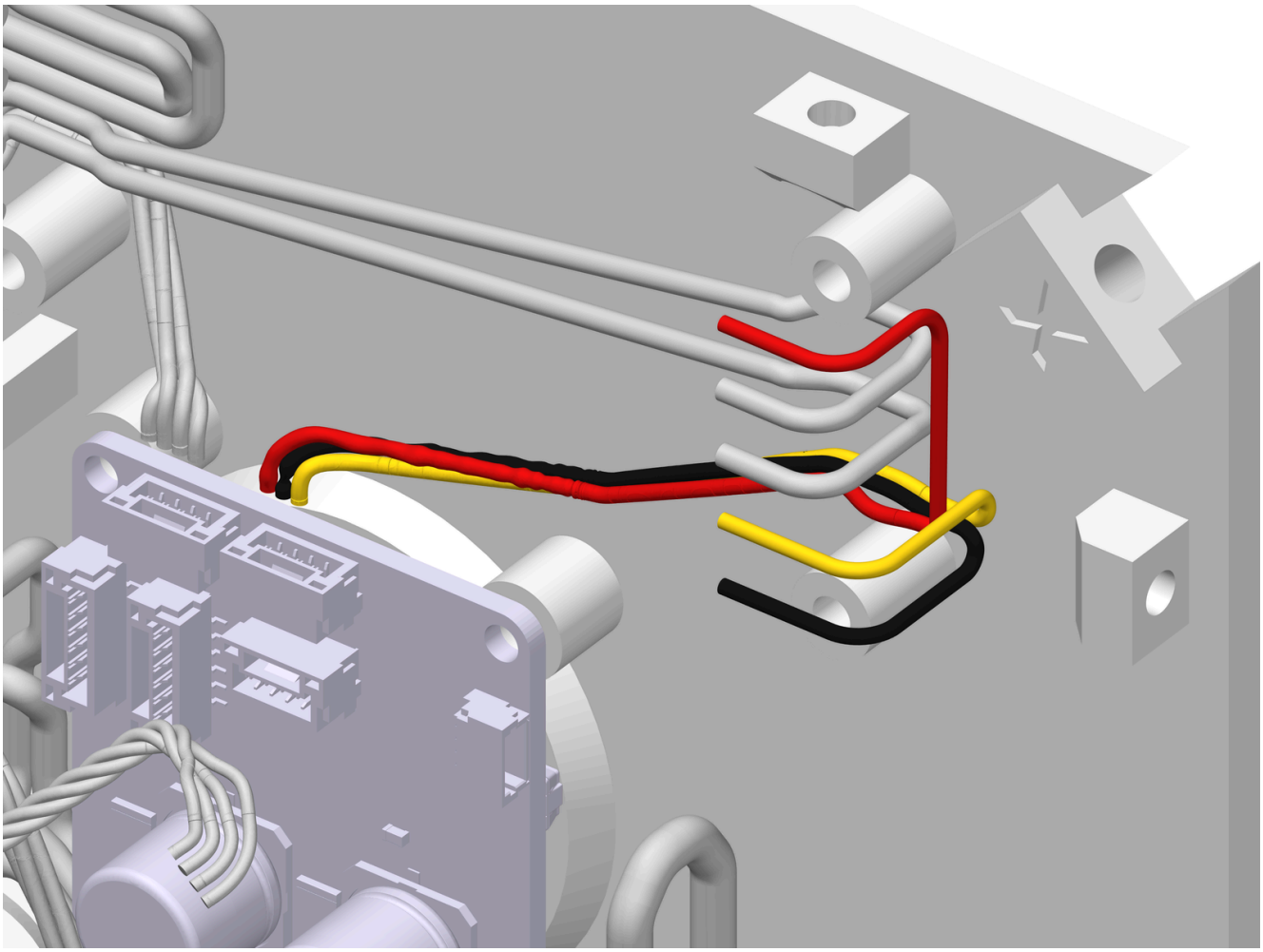
The cable goes through its round opening of clamp_y.

CAN 2 · driver X -> driver Y



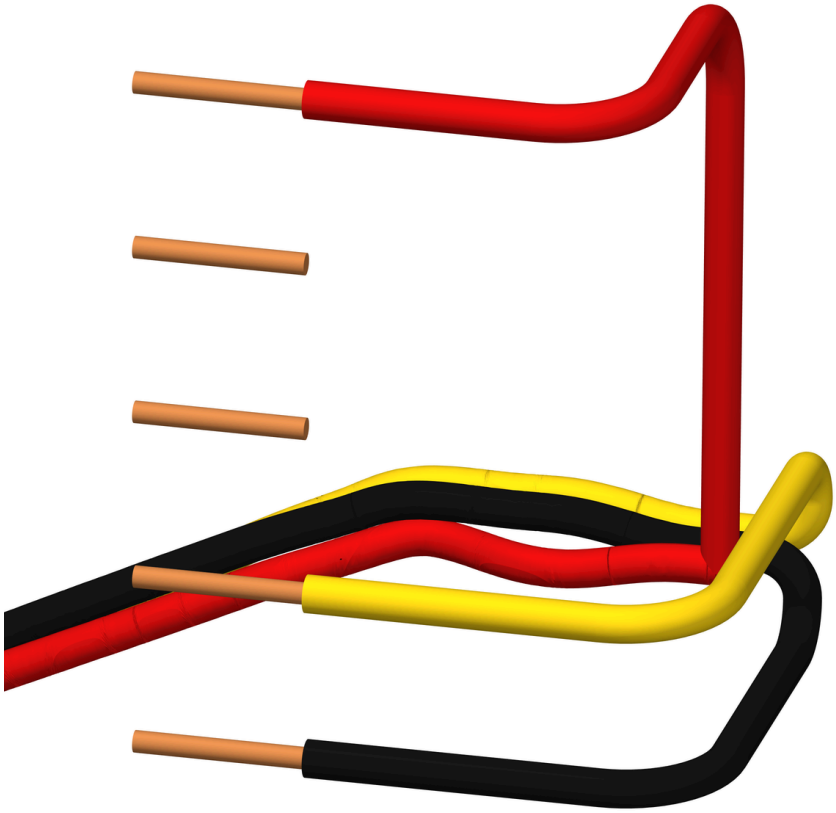
Driver X (port 13) -> driver Y (port 13).
Plugs stay loose for now.

CAN 1 · driver X -> board X



Driver X (second plug) -> board X: the cable round through clamp_x_2, then 3 wires up: yellow CANH x8, black CANL x9, red GND x5.
Plugs stay loose for now.

CAN 1 at board X



- Red -> GND (x5), yellow -> CANH (x8), black -> CANL (x9) - each on its 3 cm tail of solid wire (solder, heat-shrink), the tail into the terminal.
- At the driver, printed under each CAN port: GND (red), L (black), H (yellow), NC (white).
- The white one: out of the plug at driver X, at its root - lift the latch over its pin with a needle and pull the white wire out, pin and all. Peel it off the whole cable. No pin to lift? Cut it right at the back of the plug, a drop of glue over the cut.
- Why? At the driver the white sits on NC - not connected, it carries nothing.

CAN: the two ends

A CAN bus needs 120 Ω at each end, between CANH and CANL. Not one more, not one less.

- The chain: board X (CAN transceiver) -> driver X -> driver Y -> driver Z.
- The two ends: the transceiver and driver Z.
- The transceiver has its own resistor. Check: CANH to CANL on it, alone - about 120 Ω. If it reads kilo-ohms, it has none: solder a 120 Ω resistor across its CANH and CANL pins.
- The drivers have none: CANH to CANL on one, alone - kilo-ohms. So driver Z needs a termination plug (next page).
- Ports 13 and 14 of a controller are joined pin to pin, all but NC - that is how the bus goes through. Check it: red to red, black to black, yellow to yellow - each pair joined. White to white: nothing (NC).

on Ω

no power

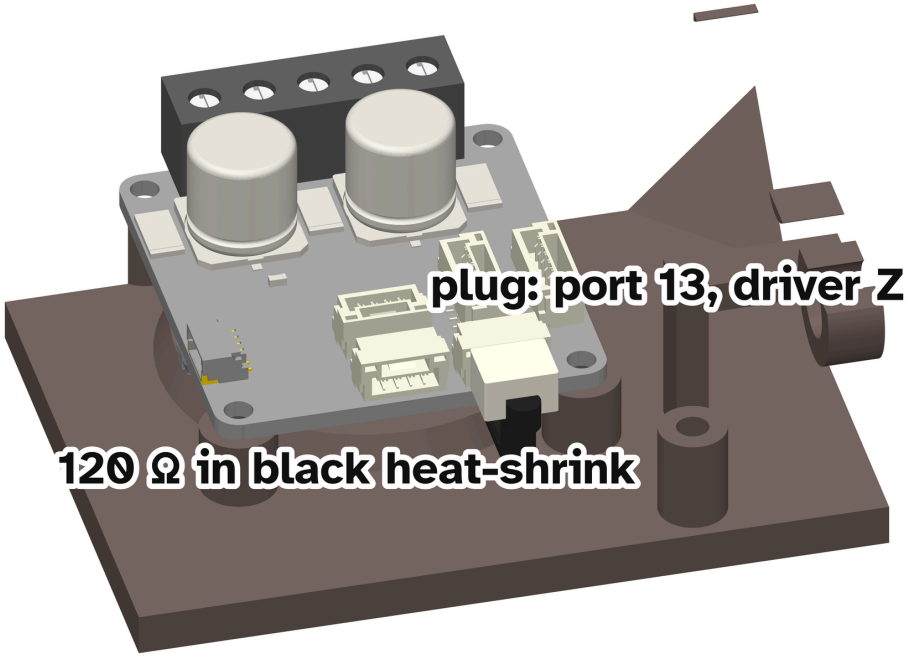
about 60 Ω

The whole bus, no power: CANH to CANL

it reads	it means
about 60 Ω	right: both ends are there
about 120 Ω	one end is missing
about 40 Ω	a third resistor: find it, take it out
beep (0 Ω)	a short circuit: find it before power

The termination plug

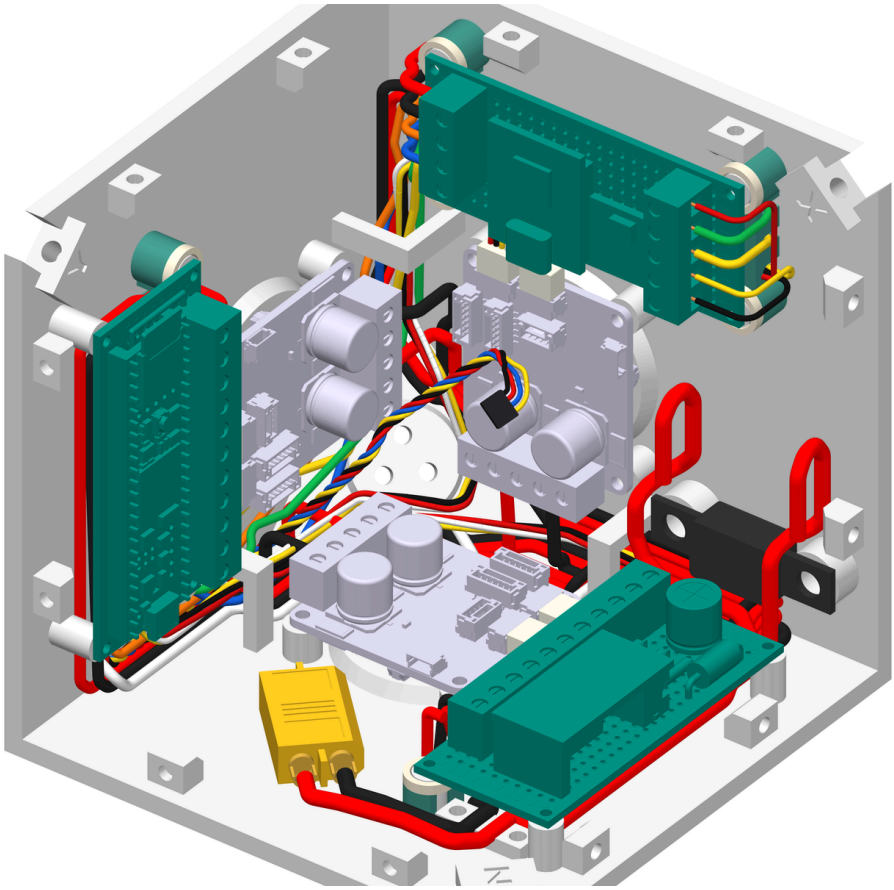
For port 13 of driver Z.



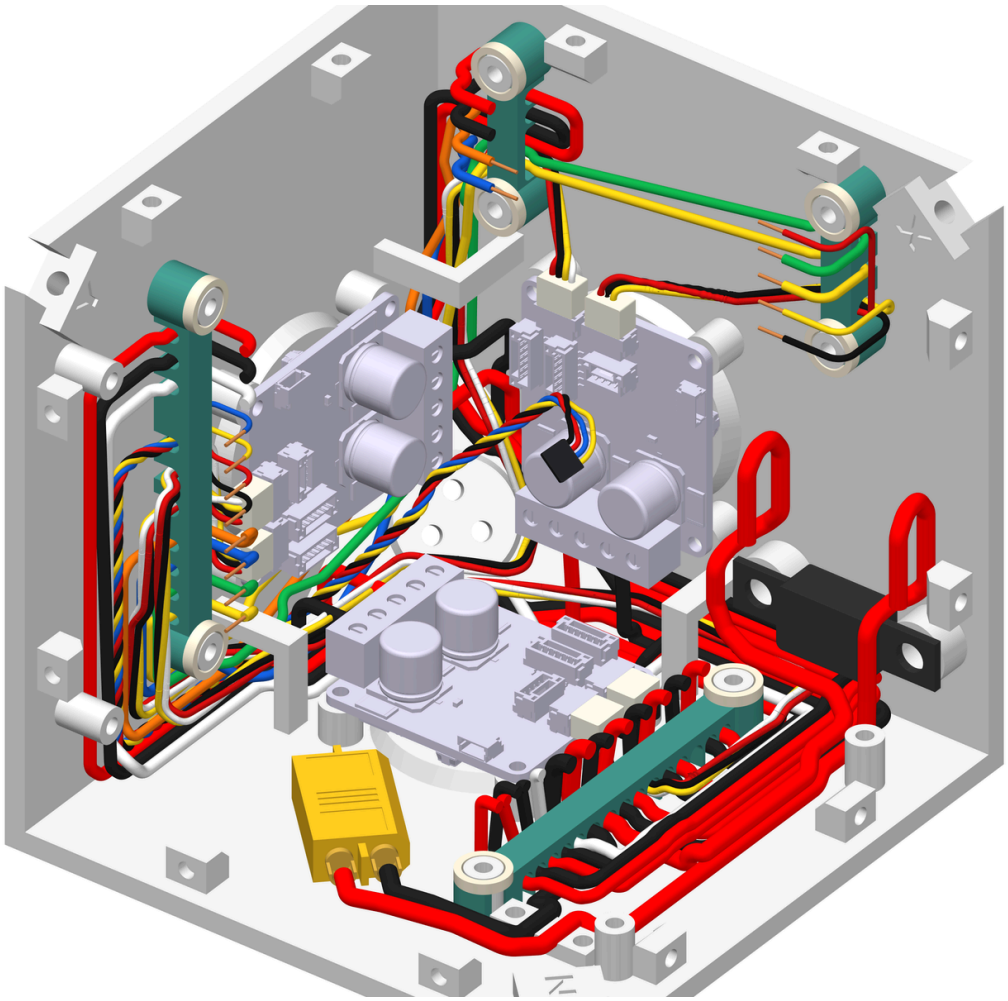
- A 1.25 mm 4-pin cable: cut it 3 cm from the plug.
- Take the red and the white out of the plug at their root: lift each latch with a needle, pull the wire out with its pin. They are not used.
- A 120 Ω resistor (brown, red, brown, gold) between the yellow (CANH) and the black (CANL). Solder, heat-shrink over each joint.
- One black heat-shrink over it all.
- Into port 13 of driver Z. Lay its tail flat under board Z.
- Check the plug alone: yellow pin to black pin - about 120 Ω.
- Check: the whole bus, no power, CANH to CANL (x8 to x9) - about 60 Ω.

5 · Boards on, connect

All wires are in. Now the boards go on and every wire goes into its terminal.



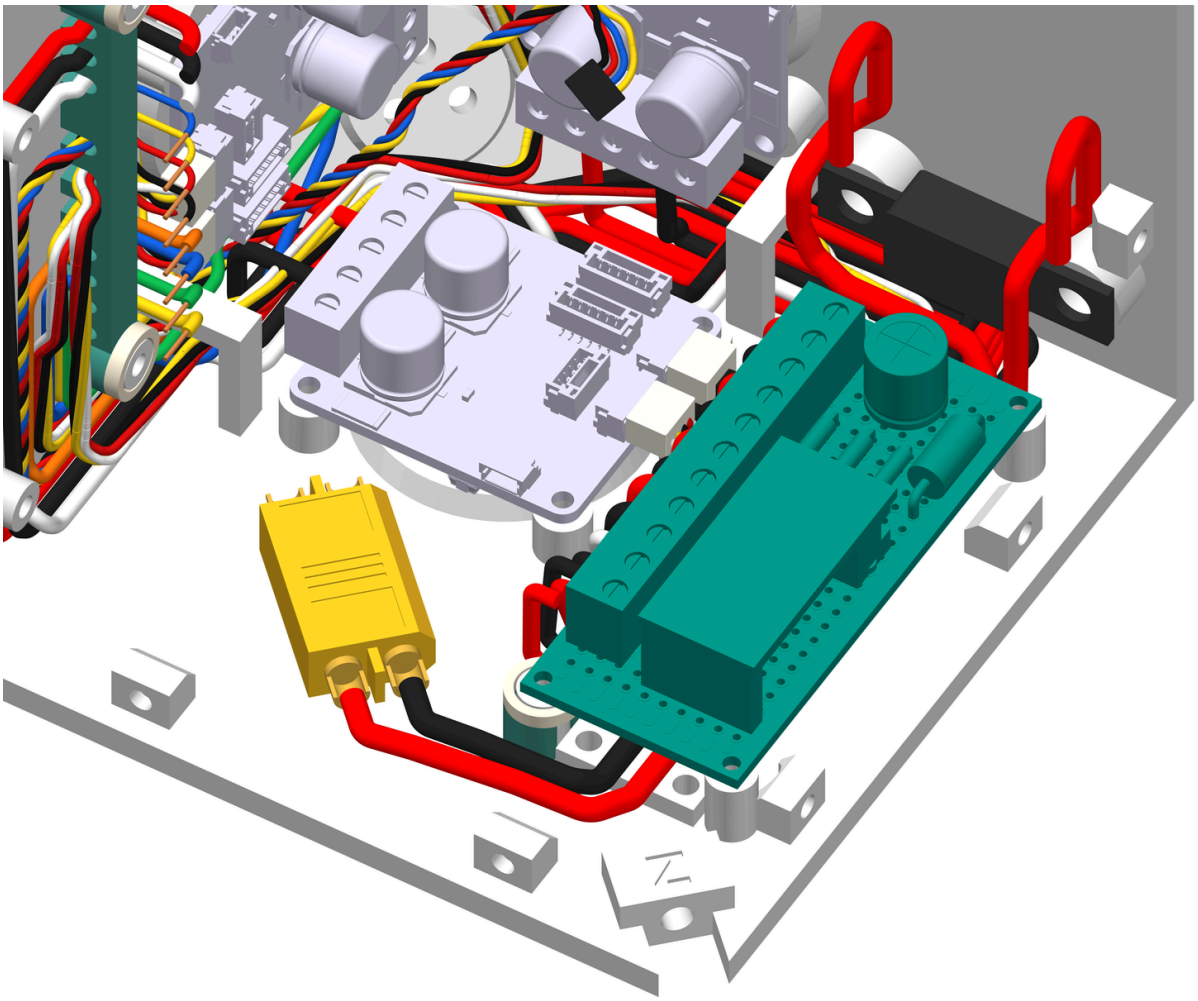
Clamps on



Each clamp over its wires: every wire under its own arch, the rings down over the board standoffs. Then a spacer on each ring - 8 in all.

Never press a wire under a clamp edge: lift the clamp and set the wire in its arch.

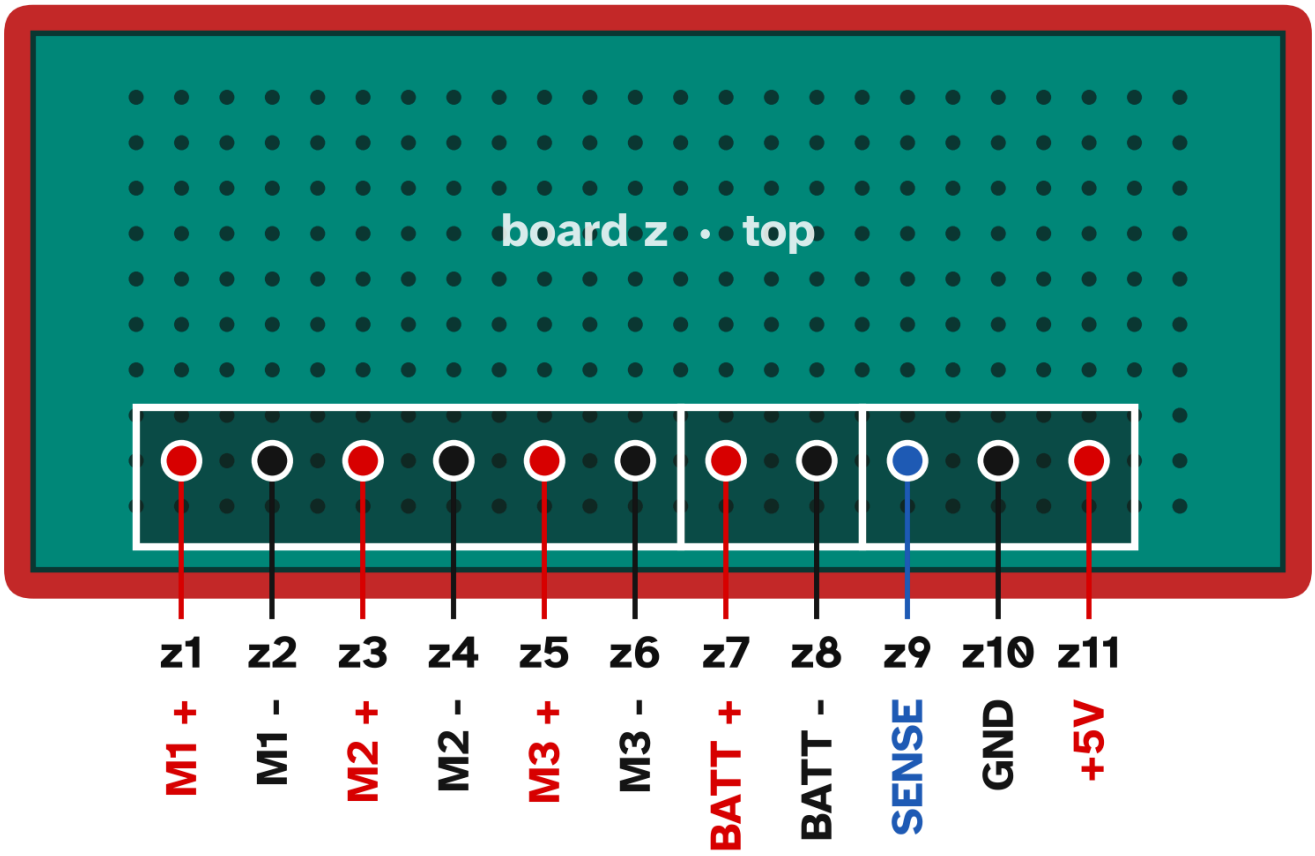
Board on



Board on its 4 standoffs - it sits on the spacers. Board Z first, then Y, then X.

Each wire end now stands in front of its terminal. Do not connect yet.

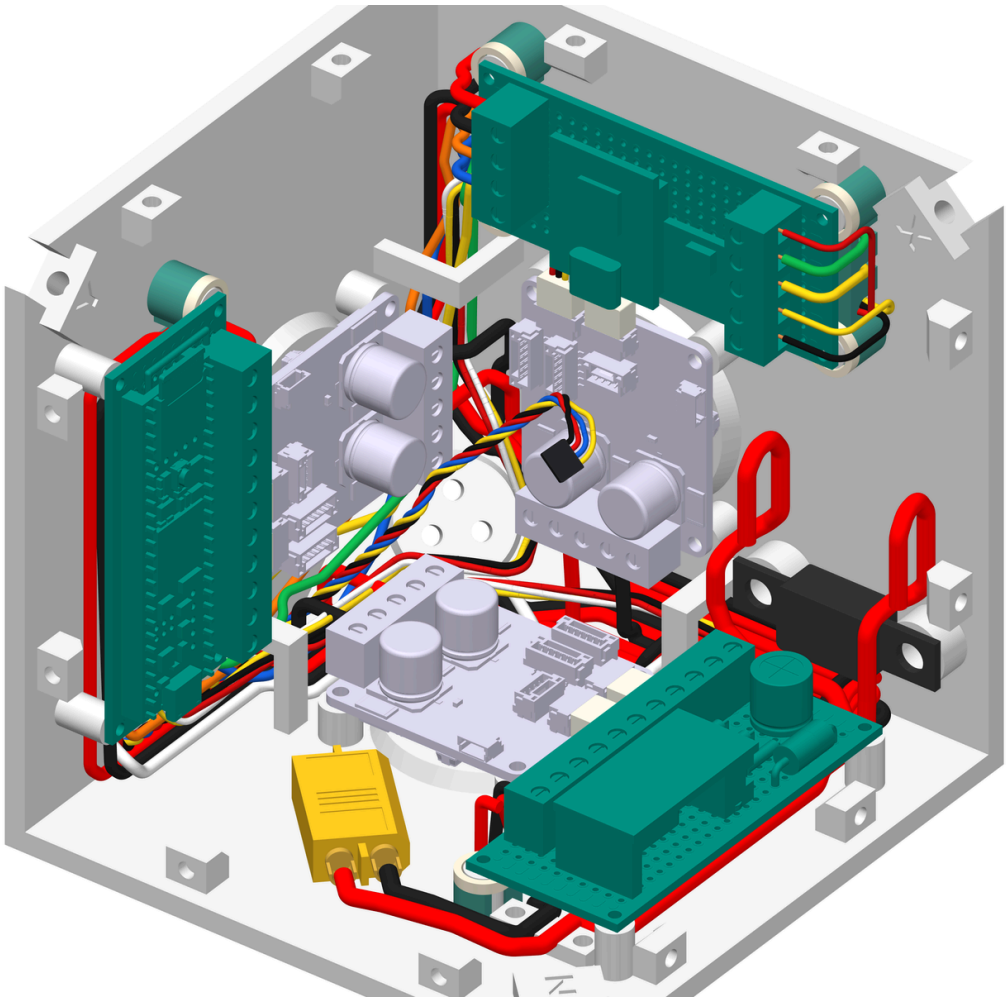
Wires into the terminals



Every wire end: mark, cut, strip 7 mm, into the pole of its name - step by step in the Assembly guide, chapter 3 (from 'Mark and cut' to 'Board X terminals').

The terminal maps are at the start of this guide (pages 'The terminals').

Plugs in



CAN plugs into the drivers, the I2C plug into the IMU.
Straight in until it clicks. Never pull a plug by its wires.

Final check

Multimeter on beep. Still NO battery.

Every wire

One probe on each end - beep. Tick it on the cut lists.

Battery connector + to -

NO beep.

Boards Z, Y, X

+5V to GND - NO beep.

CAN

CANH to CANL (x8 to x9) - about 60 Ω. CANL to GND (x9 to x5): NO beep.

Every terminal

No copper outside, no loose strand.

Only now: the first power-up

With your teacher. Battery in, watch and smell. Anything gets hot - unplug the battery at once.